part_a_classical

January 20, 2026

1 Classical Machine Learning for Video Actions

This script implements and compares three classical machine learning algorithms using extracted video features:

1. Support Vector Machine (Linear + RBF)
2. Random Forest Classifier
3. k-Nearest Neighbors (k-NN)

Features are extracted using existing project modules: - data_loader.py - feature_extraction.py

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[3]:
```
pip install -r requirements.txt
```

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Requirement already satisfied: numpy>=1.23 in /opt/conda/lib/python3.11/site-
packages (from -r requirements.txt (line 23)) (1.24.4)
Requirement already satisfied: scipy>=1.10 in /opt/conda/lib/python3.11/site-
packages (from -r requirements.txt (line 31)) (1.11.3)
Requirement already satisfied: torch>=2.0 in /opt/conda/lib/python3.11/site-
packages (from -r requirements.txt (line 42)) (2.3.0+cu121)
Requirement already satisfied: torchvision>=0.15 in
/opt/conda/lib/python3.11/site-packages (from -r requirements.txt (line 49))
(0.18.0+cu121)
Collecting openpyxl>=3.1.0 (from -r requirements.txt (line 54))
  Using cached openpyxl-3.1.5-py2.py3-none-any.whl.metadata (2.5 kB)
Collecting opencv-python-headless (from -r requirements.txt (line 69))
  Using cached
opencv_python_headless-4.13.0.90-cp37-abi3-manylinux_2_28_x86_64.whl.metadata
(19 kB)
Requirement already satisfied: scikit-learn>=1.3 in
/opt/conda/lib/python3.11/site-packages (from -r requirements.txt (line 77))
(1.3.2)
Requirement already satisfied: scikit-image in /opt/conda/lib/python3.11/site-
packages (from -r requirements.txt (line 84)) (0.22.0)
Requirement already satisfied: matplotlib>=3.7 in
/opt/conda/lib/python3.11/site-packages (from -r requirements.txt (line 96))
(3.8.4)
Requirement already satisfied: seaborn>=0.13 in /opt/conda/lib/python3.11/site-
packages (from -r requirements.txt (line 103)) (0.13.2)
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Collecting umap-learn (from -r requirements.txt (line 110))
  Using cached umap_learn-0.5.11-py3-none-any.whl.metadata (26 kB)
Requirement already satisfied: tqdm>=4.66 in /opt/conda/lib/python3.11/site-
packages (from -r requirements.txt (line 122)) (4.66.4)
Requirement already satisfied: filelock in /opt/conda/lib/python3.11/site-
packages (from torch>=2.0->-r requirements.txt (line 42)) (3.13.1)
Requirement already satisfied: typing-extensions>=4.8.0 in
/opt/conda/lib/python3.11/site-packages (from torch>=2.0->-r requirements.txt
(line 42)) (4.12.2)
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(from torch>=2.0->-r requirements.txt (line 42)) (2024.2.0)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.1.105 in
/opt/conda/lib/python3.11/site-packages (from torch>=2.0->-r requirements.txt
(line 42)) (12.1.105)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.1.105 in
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(line 42)) (12.1.105)
Requirement already satisfied: nvidia-cudnn-cu12==8.9.2.26 in
/opt/conda/lib/python3.11/site-packages (from torch>=2.0->-r requirements.txt
(line 42)) (8.9.2.26)
Requirement already satisfied: nvidia-cublas-cu12==12.1.3.1 in
/opt/conda/lib/python3.11/site-packages (from torch>=2.0->-r requirements.txt
(line 42)) (12.1.3.1)
Requirement already satisfied: nvidia-cufft-cu12==11.0.2.54 in
/opt/conda/lib/python3.11/site-packages (from torch>=2.0->-r requirements.txt
(line 42)) (11.0.2.54)
Requirement already satisfied: nvidia-curand-cu12==10.3.2.106 in
/opt/conda/lib/python3.11/site-packages (from torch>=2.0->-r requirements.txt
(line 42)) (10.3.2.106)
Requirement already satisfied: nvidia-cusolver-cu12==11.4.5.107 in
/opt/conda/lib/python3.11/site-packages (from torch>=2.0->-r requirements.txt
(line 42)) (11.4.5.107)
Requirement already satisfied: nvidia-cusparse-cu12==12.1.0.106 in
/opt/conda/lib/python3.11/site-packages (from torch>=2.0->-r requirements.txt
(line 42)) (12.1.0.106)
Requirement already satisfied: nvidia-nccl-cu12==2.20.5 in
/opt/conda/lib/python3.11/site-packages (from torch>=2.0->-r requirements.txt
(line 42)) (2.20.5)
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/opt/conda/lib/python3.11/site-packages (from nvidia-cusolver-
cu12==11.4.5.107->torch>=2.0->-r requirements.txt (line 42)) (12.1.105)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in
/opt/conda/lib/python3.11/site-packages (from torchvision>=0.15->-r
requirements.txt (line 49)) (10.4.0)
Collecting et-xmlfile (from openpyxl>=3.1.0->-r requirements.txt (line 54))
  Using cached et_xmlfile-2.0.0-py3-none-any.whl.metadata (2.7 kB)
INFO: pip is looking at multiple versions of opencv-python-headless to determine
which version is compatible with other requirements. This could take a while.
Collecting opencv-python-headless (from -r requirements.txt (line 69))
  Using cached opencv_python_headless-4.12.0.88-cp37-abi3-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (19 kB)
  Using cached opencv_python_headless-4.11.0.86-cp37-abi3-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (20 kB)
Requirement already satisfied: joblib>=1.1.1 in /opt/conda/lib/python3.11/site-
packages (from scikit-learn>=1.3->-r requirements.txt (line 77)) (1.4.2)
Requirement already satisfied: threadpoolctl>=2.0.0 in
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requirements.txt (line 77)) (3.5.0)
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(line 84)) (2024.7.2)
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(line 84)) (0.4)
Requirement already satisfied: contourpy>=1.0.1 in
/opt/conda/lib/python3.11/site-packages (from matplotlib>=3.7->-r
requirements.txt (line 96)) (1.2.1)
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Requirement already satisfied: fonttools>=4.22.0 in
/opt/conda/lib/python3.11/site-packages (from matplotlib>=3.7->-r
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Requirement already satisfied: kiwisolver>=1.3.1 in
/opt/conda/lib/python3.11/site-packages (from matplotlib>=3.7->-r
requirements.txt (line 96)) (1.4.5)
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/opt/conda/lib/python3.11/site-packages (from matplotlib>=3.7->-r
requirements.txt (line 96)) (3.1.2)
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Requirement already satisfied: python-dateutil>=2.7 in
/opt/conda/lib/python3.11/site-packages (from matplotlib>=3.7->-r
requirements.txt (line 96)) (2.9.0)
Requirement already satisfied: pandas>=1.2 in /opt/conda/lib/python3.11/site-
packages (from seaborn>=0.13->-r requirements.txt (line 103)) (2.1.4)
Collecting scikit-learn>=1.3 (from -r requirements.txt (line 77))
  Downloading scikit_learn-1.8.0-cp311-cp311-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (11 kB)
Collecting numba>=0.51.2 (from umap-learn->-r requirements.txt (line 110))
  Downloading
numba-0.63.1-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata
(2.9 kB)
Collecting pynndescent>=0.5 (from umap-learn->-r requirements.txt (line 110))
  Downloading pynndescent-0.6.0-py3-none-any.whl.metadata (6.9 kB)
Collecting llvmlite<0.47,>=0.46.0dev0 (from numba>=0.51.2->umap-learn->-r
requirements.txt (line 110))
  Downloading llvmlite-0.46.0-cp311-cp311-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (5.0 kB)
Requirement already satisfied: pytz>=2020.1 in /opt/conda/lib/python3.11/site-
packages (from pandas>=1.2->seaborn>=0.13->-r requirements.txt (line 103))
(2024.1)
Requirement already satisfied: tzdata>=2022.1 in /opt/conda/lib/python3.11/site-
packages (from pandas>=1.2->seaborn>=0.13->-r requirements.txt (line 103))
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Requirement already satisfied: six>=1.5 in /opt/conda/lib/python3.11/site-
packages (from python-dateutil>=2.7->matplotlib>=3.7->-r requirements.txt (line
96)) (1.16.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/opt/conda/lib/python3.11/site-packages (from jinja2->torch>=2.0->-r
requirements.txt (line 42)) (2.1.5)
Requirement already satisfied: mpmath>=0.19 in /opt/conda/lib/python3.11/site-
packages (from sympy->torch>=2.0->-r requirements.txt (line 42)) (1.3.0)
Downloading openpyxl-3.1.5-py2.py3-none-any.whl (250 kB)
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Downloading opencv_python_headless-4.11.0.86-cp37-abi3-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl (50.0 MB)
                         50.0/50.0 MB
1.2 MB/s eta 0:00:0000:010m00:02
Downloading umap_learn-0.5.11-py3-none-any.whl (90 kB)
                         90.9/90.9 kB
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scikit_learn-1.8.0-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl
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llvmlite-0.46.0-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (56.3
MB)
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2.1 MB/s eta 0:00:0000:0100:01
Installing collected packages: opencv-python-headless, llvmlite, et-
xmlfile, scikit-learn, openpyxl, numba, pynndescent, umap-learn
  Attempting uninstall: scikit-learn
    Found existing installation: scikit-learn 1.3.2
    Uninstalling scikit-learn-1.3.2:
      Successfully uninstalled scikit-learn-1.3.2
Successfully installed et-xmlfile-2.0.0 llvmlite-0.46.0 numba-0.63.1 opencv-
python-headless-4.11.0.86 openpyxl-3.1.5 pynndescent-0.6.0 scikit-learn-1.8.0
umap-learn-0.5.11
Note: you may need to restart the kernel to use updated packages.
```

[1]:
```python
# ==========================
# Standard Library Imports
# ==========================
import os
import time
from typing import Dict, Tuple
import warnings

# ==========================
# Third-Party Imports
# ==========================
import numpy as np
import pandas as pd
from pathlib import Path
# Object-oriented, cross-platform file system path handling
import matplotlib.pyplot as plt

from evaluate import evaluate_classification

from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import (
    accuracy_score,
```

```
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix,
    ConfusionMatrixDisplay,
)

from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.neighbors import KNeighborsClassifier
from IPython.display import display, Markdown

# ========================
# Project Imports
# ========================
from feature_extraction import (
    visualize_classical_feature_importance,
    visualize_classical_feature_space,
    get_classical_feature_names
)
from data_loader import load_split_data_and_extract_features
from evaluate import create_full_comparison_table, generate_dynamic_observations
```

[2]:
```
# ========================
# Global Configurations
# ========================
warnings.filterwarnings("ignore")
RANDOM_STATE = 42
CV_FOLDS = 5


# ================================================
# Utility Functions
# ================================================
def evaluate_model(
    model,
    X_test: np.ndarray,
    y_test: np.ndarray,
) -> Dict[str, float]:
    """
    Evaluate a trained model using multiple metrics.

    Returns:
        Dictionary containing accuracy, precision, recall, and F1-score.
    """
    y_pred = model.predict(X_test)
```

```
    return {
        "accuracy": accuracy_score(y_test, y_pred),
        "precision": precision_score(y_test, y_pred, average="macro"),
        "recall": recall_score(y_test, y_pred, average="macro"),
        "f1_score": f1_score(y_test, y_pred, average="macro"),
    }


def plot_confusion_matrix(model, X_test, y_test, title: str) -> None:
    """
    Plot confusion matrix for a trained classifier.
    """
    cm = confusion_matrix(y_test, model.predict(X_test))
    disp = ConfusionMatrixDisplay(confusion_matrix=cm)
    disp.plot()
    plt.title(title)
    plt.show()
```

[3]:
```
# ================================================
# Data Loading & Feature Extraction
# ================================================
print("\n Loading dataset splits and extract features...")

print("\n--------------------------------------------------------------------------")
print("* Processing training dataset.....")
print("--------------------------------------------------------------------------\n")
X_train, y_train, CLASS_TO_LABEL, LABEL_TO_CLASS =␣
 ↪load_split_data_and_extract_features("train")

print("\n--------------------------------------------------------------------------")
print("* Processing validation dataset.....")
print("--------------------------------------------------------------------------\n")
X_val, y_val, _, _ = load_split_data_and_extract_features(split_name="val")

print("\n--------------------------------------------------------------------------")
print("* Processing test dataset.....")
print("--------------------------------------------------------------------------\n")
X_test, y_test, _, _ = load_split_data_and_extract_features(split_name="test")

print(f"\n\n* Train shape: {X_train.shape}")
print(f"* Test shape : {X_test.shape}")
```

```
 Loading dataset splits and extract features…

--------------------------------------------------------------------------
* Processing training dataset…
```

```
------------------------------------------------------------------

 Detected classes (5): ['class_1_Diving', 'class_2_JumpRope',
'class_3_SkyDiving', 'class_4_Surfing', 'class_5_Swing']

[INFO] * processing Video file : class_1_Diving/v_Diving_g12_c02.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g09_c04.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g10_c05.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g19_c02.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g14_c04.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g15_c06.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g17_c02.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g02_c05.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g16_c05.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g06_c03.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g18_c03.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g09_c03.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g22_c06.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g06_c01.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g03_c04.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g02_c04.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g12_c04.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g16_c01.avi
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[INFO] * processing Video file : class_1_Diving/v_Diving_g19_c04.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g04_c05.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g15_c01.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g13_c07.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g21_c03.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g02_c06.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g01_c05.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g12_c03.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g24_c06.avi
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[INFO] * processing Video file : class_1_Diving/v_Diving_g12_c07.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g13_c02.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g21_c04.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g23_c04.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g17_c01.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g06_c06.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g16_c03.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g07_c03.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g03_c02.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g16_c02.avi
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[INFO] * processing Video file : class_1_Diving/v_Diving_g08_c05.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g07_c04.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g21_c06.avi
```

```
[INFO] * processing Video file : class_1_Diving/v_Diving_g10_c06.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g02_c02.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g18_c01.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g12_c06.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g15_c05.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g01_c07.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g08_c03.avi
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[INFO] * processing Video file : class_1_Diving/v_Diving_g23_c06.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g01_c02.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g07_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g07_c01.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g03_c04.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g22_c03.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g16_c01.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g05_c01.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g24_c01.avi
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[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g19_c03.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g03_c01.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g08_c02.avi
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[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g11_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g04_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g10_c03.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g10_c01.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g21_c03.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g05_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g24_c04.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g20_c01.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g23_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g02_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g14_c04.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g05_c03.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g17_c03.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g08_c01.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g22_c01.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g04_c03.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g20_c04.avi
```

```
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g22_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g23_c01.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g25_c03.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g06_c04.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g25_c04.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g14_c01.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g17_c05.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g15_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g17_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g12_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g07_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g18_c05.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g09_c04.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g01_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g24_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g01_c03.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g16_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g08_c04.avi
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[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g21_c04.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g17_c04.avi
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[INFO] * processing Video file : class_4_Surfing/v_Surfing_g16_c02.avi
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[INFO] * processing Video file : class_4_Surfing/v_Surfing_g02_c04.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g16_c01.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g14_c04.avi
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[INFO] * processing Video file : class_4_Surfing/v_Surfing_g15_c02.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g17_c06.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g04_c04.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g25_c02.avi
```

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[INFO] * processing Video file : class_4_Surfing/v_Surfing_g16_c06.avi
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[INFO] * processing Video file : class_4_Surfing/v_Surfing_g20_c02.avi
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[INFO] * processing Video file : class_4_Surfing/v_Surfing_g09_c03.avi
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[INFO] * processing Video file : class_4_Surfing/v_Surfing_g01_c03.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g10_c05.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g10_c06.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g20_c05.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g17_c02.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g01_c05.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g02_c05.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g03_c02.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g08_c02.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g07_c05.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g19_c02.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g09_c06.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g12_c03.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g21_c02.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g22_c02.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g17_c01.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g11_c06.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g15_c05.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g13_c04.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g09_c05.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g14_c04.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g06_c02.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g09_c03.avi
```

```
[INFO] * processing Video file : class_5_Swing/v_Swing_g14_c03.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g07_c02.avi
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[INFO] * processing Video file : class_5_Swing/v_Swing_g18_c04.avi
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[INFO] * processing Video file : class_5_Swing/v_Swing_g03_c04.avi
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[INFO] * processing Video file : class_5_Swing/v_Swing_g08_c05.avi
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[INFO] * processing Video file : class_5_Swing/v_Swing_g23_c03.avi
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[INFO] * processing Video file : class_5_Swing/v_Swing_g02_c03.avi
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[INFO] * processing Video file : class_5_Swing/v_Swing_g20_c05.avi
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[INFO] * processing Video file : class_5_Swing/v_Swing_g04_c06.avi
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[INFO] * processing Video file : class_5_Swing/v_Swing_g06_c01.avi
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[INFO] * processing Video file : class_5_Swing/v_Swing_g10_c02.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g15_c07.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g07_c06.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g13_c04.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g13_c01.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g14_c01.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g17_c03.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g24_c01.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g12_c04.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g12_c02.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g24_c06.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g22_c01.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g16_c06.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g10_c01.avi
```

```
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g17_c02.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g20_c03.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g05_c03.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g23_c05.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g11_c02.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g25_c05.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g24_c03.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g25_c06.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g15_c01.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g04_c01.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g14_c04.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g20_c04.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g24_c05.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g11_c07.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g08_c02.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g16_c07.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g07_c06.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g03_c01.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g19_c03.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g15_c05.avi
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[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g16_c05.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g17_c01.avi
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[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g13_c03.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g15_c02.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g11_c06.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g21_c06.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g02_c01.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g04_c04.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g04_c06.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g09_c01.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g14_c02.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g15_c04.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g03_c03.avi
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[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g20_c01.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g06_c03.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g17_c05.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g03_c02.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g09_c05.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g24_c02.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g09_c03.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g12_c01.avi

--------------------------------------------------------------------
* Processing validation dataset…
```

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------------------------------------------------------------------

 Detected classes (5): ['class_1_Diving', 'class_2_JumpRope',
'class_3_SkyDiving', 'class_4_Surfing', 'class_5_Swing']

[INFO] * processing Video file : class_1_Diving/v_Diving_g07_c01.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g22_c05.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g14_c02.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g21_c05.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g06_c02.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g03_c03.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g01_c04.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g11_c05.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g01_c06.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g15_c07.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g21_c07.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g11_c06.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g14_c03.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g18_c01.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g11_c04.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g23_c04.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g05_c04.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g15_c04.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g11_c03.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g16_c03.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g06_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g07_c05.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g18_c03.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g21_c01.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g01_c07.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g25_c03.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g23_c04.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g01_c04.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g13_c01.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g07_c03.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g16_c04.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g13_c02.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g20_c04.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g18_c01.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g03_c03.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g17_c04.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g05_c07.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g21_c01.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g25_c02.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g19_c03.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g02_c01.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g08_c03.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g10_c04.avi
```

```
[INFO] * processing Video file : class_5_Swing/v_Swing_g17_c03.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g05_c05.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g23_c02.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g25_c04.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g15_c06.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g09_c06.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g02_c06.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g09_c07.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g01_c02.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g19_c04.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g09_c02.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g18_c07.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g22_c02.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g10_c04.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g05_c01.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g18_c04.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g08_c01.avi

--------------------------------------------------------------------
* Processing test dataset…
--------------------------------------------------------------------

 Detected classes (5): ['class_1_Diving', 'class_2_JumpRope',
'class_3_SkyDiving', 'class_4_Surfing', 'class_5_Swing']

[INFO] * processing Video file : class_1_Diving/v_Diving_g04_c04.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g14_c01.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g13_c03.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g24_c04.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g09_c01.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g17_c03.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g06_c04.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g08_c06.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g19_c01.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g22_c04.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g03_c05.avi
[INFO] * processing Video file : class_1_Diving/v_Diving_g13_c05.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g25_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g05_c05.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g14_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g12_c04.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g11_c01.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g06_c03.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g09_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g13_c02.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g09_c01.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g01_c01.avi
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g03_c05.avi
```

```
[INFO] * processing Video file : class_3_SkyDiving/v_SkyDiving_g25_c01.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g24_c03.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g13_c05.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g14_c02.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g17_c05.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g17_c07.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g19_c03.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g24_c04.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g04_c03.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g08_c03.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g09_c01.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g10_c01.avi
[INFO] * processing Video file : class_4_Surfing/v_Surfing_g11_c04.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g23_c04.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g24_c05.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g06_c04.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g16_c01.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g12_c02.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g14_c02.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g07_c03.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g24_c01.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g04_c07.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g13_c02.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g07_c04.avi
[INFO] * processing Video file : class_5_Swing/v_Swing_g25_c01.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g21_c04.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g19_c01.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g12_c03.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g08_c03.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g04_c07.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g12_c05.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g07_c03.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g21_c05.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g05_c05.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g22_c06.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g16_c03.avi
[INFO] * processing Video file : class_2_JumpRope/v_JumpRope_g20_c02.avi


* Train shape: (280, 399)
* Test shape : (60, 399)
```

[4]:
```
# ================================================
# 1  SUPPORT VECTOR MACHINE (LINEAR + RBF)
# ================================================
# This block trains an SVM classifier using GridSearchCV
# and performs post-training feature analysis including:
```

```
# - Feature importance (linear kernel only)
# - t-SNE visualization
# - UMAP visualization
# ================================================

print("\n---------------------------------------")
print("  Training Support Vector Machine...")
print("---------------------------------------\n")

# ------------------------------------------------
#   Step 1: Define SVM pipeline
# ------------------------------------------------
# StandardScaler is REQUIRED for SVM to:
# - normalize feature magnitudes
# - ensure stable optimization
# - improve convergence
# ------------------------------------------------

svm_pipeline = Pipeline(
    steps=[
        ("scaler", StandardScaler()),
        (
            "svm",
            SVC(
                probability=True,    # Required for ROC/AUC computation
                random_state=42
            ),
        ),
    ]
)

# ------------------------------------------------
#   Step 2: Define hyperparameter search space
# ------------------------------------------------
# - Linear kernel → supports feature importance via coefficients
# - RBF kernel → non-linear, higher capacity but no interpretability
# ------------------------------------------------

svm_param_grid = [
    {
        "svm__kernel": ["linear"],
        "svm__C": [0.1, 1, 10],
    },
    {
        "svm__kernel": ["rbf"],
        "svm__C": [0.1, 1, 10],
        "svm__gamma": [0.01, 0.1, 1],
```

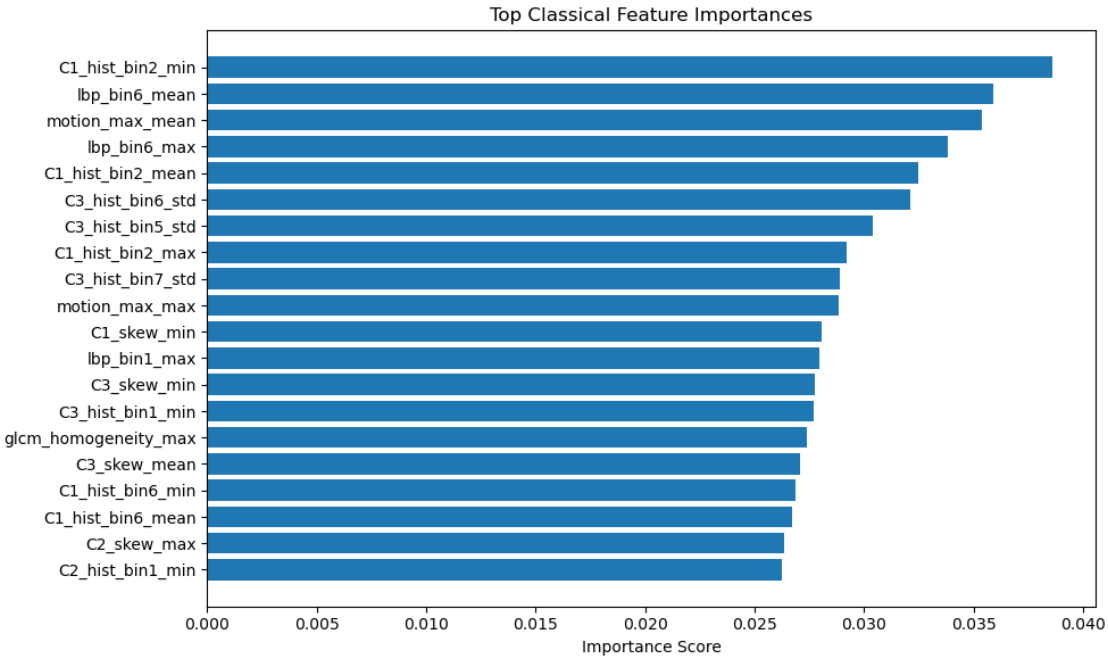
```
    },
]

# ------------------------------------------------
#   Step 3: GridSearchCV setup
# ------------------------------------------------
# - Cross-validation ensures robust hyperparameter selection
# - Accuracy is used as optimization metric
# ------------------------------------------------

svm_grid = GridSearchCV(
    estimator=svm_pipeline,
    param_grid=svm_param_grid,
    cv=CV_FOLDS,
    scoring="accuracy",
    n_jobs=-1,
    verbose=1,
)

# ------------------------------------------------
#   Step 4: Measure training time
# ------------------------------------------------

train_start_time = time.perf_counter()
svm_grid.fit(X_train, y_train)
train_end_time = time.perf_counter()

svm_training_time = train_end_time - train_start_time

print(f"\n* SVM Training Time: {svm_training_time:.4f} seconds")
print("* Best SVM Parameters:", svm_grid.best_params_)

# ------------------------------------------------
#   Step 5: Extract best trained model
# ------------------------------------------------
# best_svm is a FULL pipeline:
#   scaler → trained SVM
# ------------------------------------------------

best_svm = svm_grid.best_estimator_

# ================================================
#   CLASSICAL FEATURE ANALYSIS (POST-TRAINING)
# ================================================
# IMPORTANT:
# - Feature importance requires trained model
# - Feature space visualization must use SCALED features
```

```
# ================================================

# ------------------------------------------------
#   Step 6: Feature importance (Linear SVM only)
# ------------------------------------------------
# Only linear SVM exposes coefficients (coef_)
# ------------------------------------------------

feature_names = get_classical_feature_names(
    hist_bins=16,
    lbp_points=8,
    include_motion=True
)

print(f"\n  Total classical features: {len(feature_names)}\n")

if best_svm.named_steps["svm"].kernel == "linear":
    visualize_classical_feature_importance(
        model=best_svm.named_steps["svm"],    # trained linear SVM
        feature_names=feature_names,          # handcrafted feature names
        top_k=20,
        save_name="svm_feature_importance.png"
    )
else:
    print("\n  Feature importance skipped (RBF kernel selected)")

# ------------------------------------------------
#   Step 7: Extract scaled features for visualization
# ------------------------------------------------
# Feature space must match what the SVM actually learned
# ------------------------------------------------

X_train_scaled = best_svm.named_steps["scaler"].transform(X_train)

# ------------------------------------------------
#   Step 8: t-SNE visualization
# ------------------------------------------------
# Shows how samples cluster in reduced 2D space
# ------------------------------------------------

visualize_classical_feature_space(
    features=X_train_scaled,
    labels=y_train,
    method="tsne",
    save_name="svm_tsne.png"
)
```

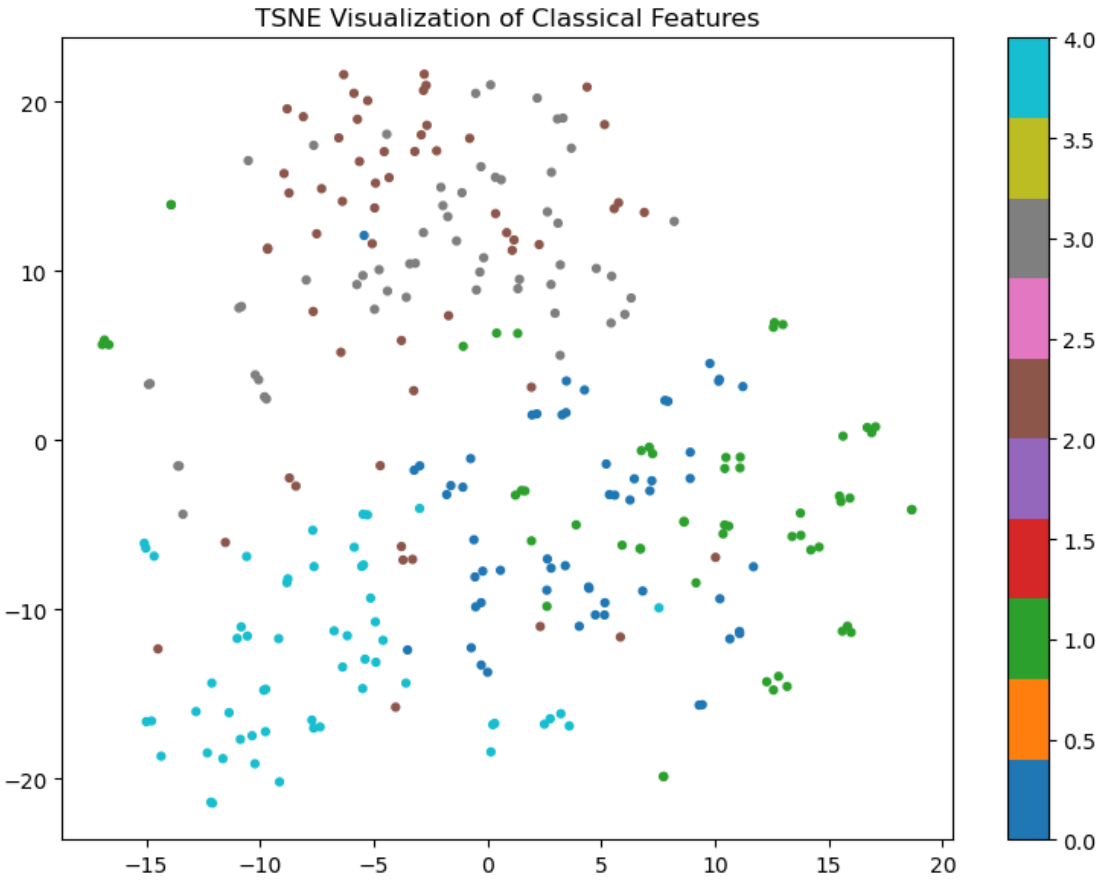
```
# ------------------------------------------------
#   Step 9: UMAP visualization
# ------------------------------------------------
# Preserves both local and global structure
# ------------------------------------------------

visualize_classical_feature_space(
    features=X_train_scaled,
    labels=y_train,
    method="umap",
    save_name="svm_umap.png"
)

# ================================================
#   INFERENCE TIME PER VIDEO
# ================================================
# Measures average prediction latency
# ================================================

inference_start_time = time.perf_counter()
y_pred = best_svm.predict(X_test)
inference_end_time = time.perf_counter()

svm_inference_time_per_video = (
    (inference_end_time - inference_start_time) / len(X_test)
)

print(
    f"  Avg Inference Time per Video: "
    f"{svm_inference_time_per_video:.6f} seconds\n"
)

# ================================================
# ROC / AUC PROBABILITIES
# ================================================
# probability=True enables predict_proba()
# ================================================

y_scores = best_svm.predict_proba(X_test)

# ================================================
# UNIFIED EVALUATION
# ================================================

class_names = [
    LABEL_TO_CLASS[i].replace("class_", "")
    for i in range(len(LABEL_TO_CLASS))
```

```
]
svm_metrics = evaluate_classification(
    y_true=y_test,
    y_pred=y_pred,
    y_scores=y_scores,
    class_names=class_names,
    model_name="svm"
)

# Store efficiency metrics for later comparison
svm_metrics["training_time"] = svm_training_time
svm_metrics["inference_time_per_video"] = svm_inference_time_per_video

print("\n SVM Evaluation Completed")
```

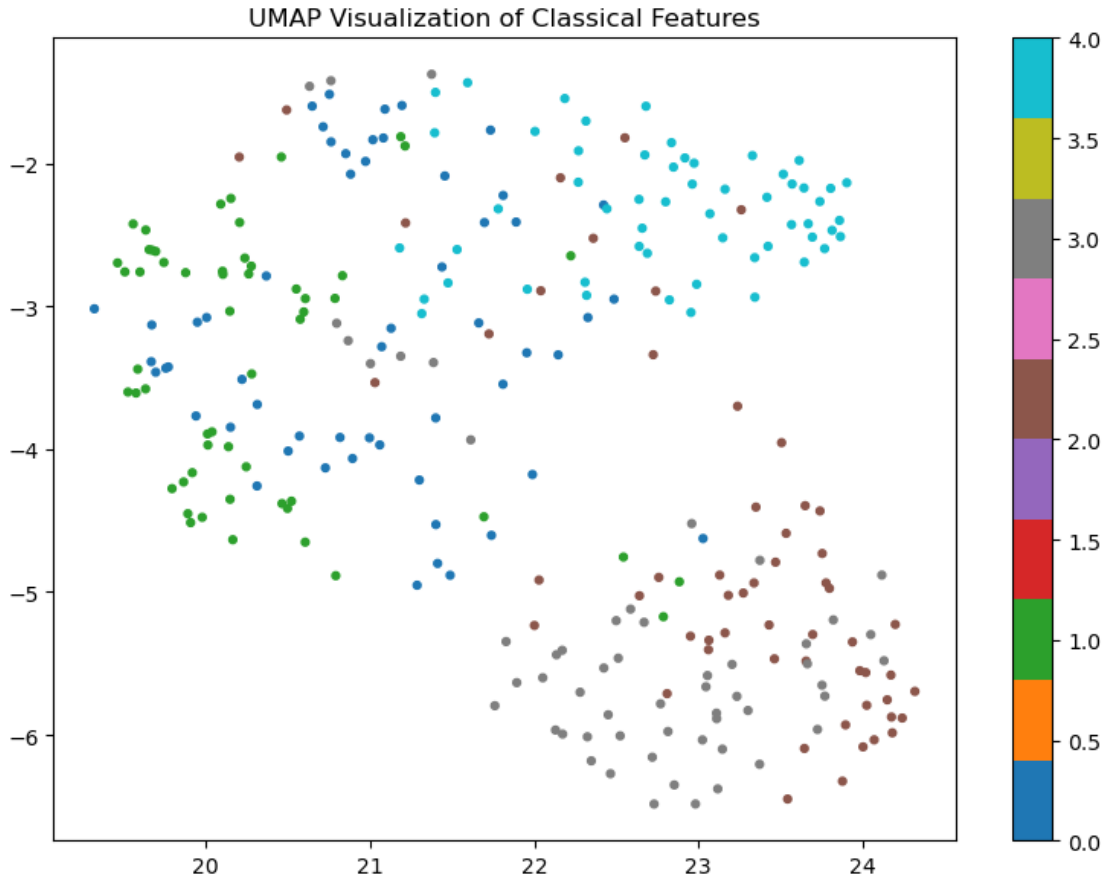
```
--------------------------------------
 Training Support Vector Machine…
--------------------------------------

Fitting 5 folds for each of 12 candidates, totalling 60 fits

* SVM Training Time: 1.5324 seconds
* Best SVM Parameters: {'svm__C': 0.1, 'svm__kernel': 'linear'}

 Total classical features: 399

[Saved] Feature importance → /home/jovyan/va-assignment-
1/Student_2024ab05275_Video_Classification/results/feature_visualizations/classi
cal/svm_feature_importance.png
```



[Saved] TSNE plot → /home/jovyan/va-assignment-1/Student_2024ab05275_Video_Classification/results/feature_visualizations/classical/svm_tsne.png



[Saved] UMAP plot → /home/jovyan/va-assignment-1/Student_2024ab05275_Video_Classification/results/feature_visualizations/classical/svm_umap.png



Avg Inference Time per Video: 0.000031 seconds

Starting evaluation for: svm

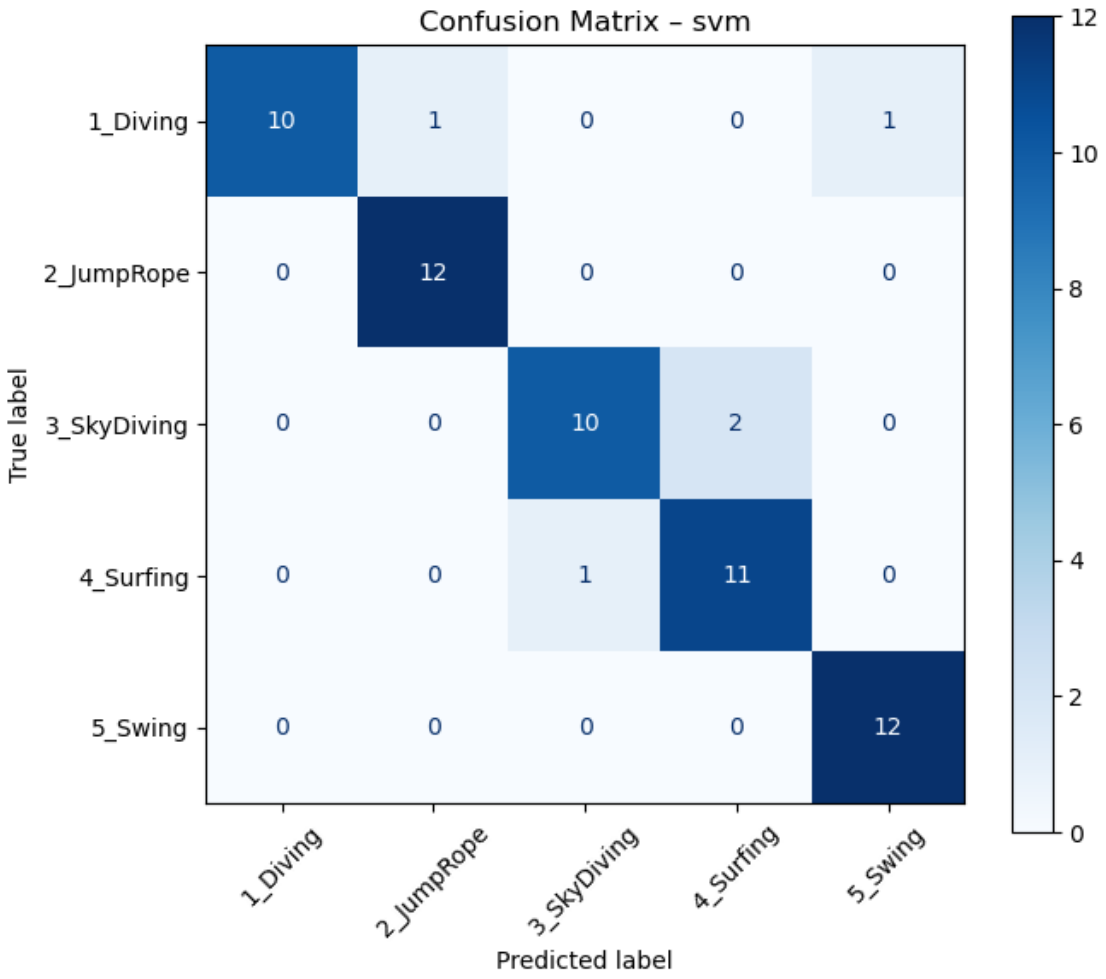
Overall Performance Metrics

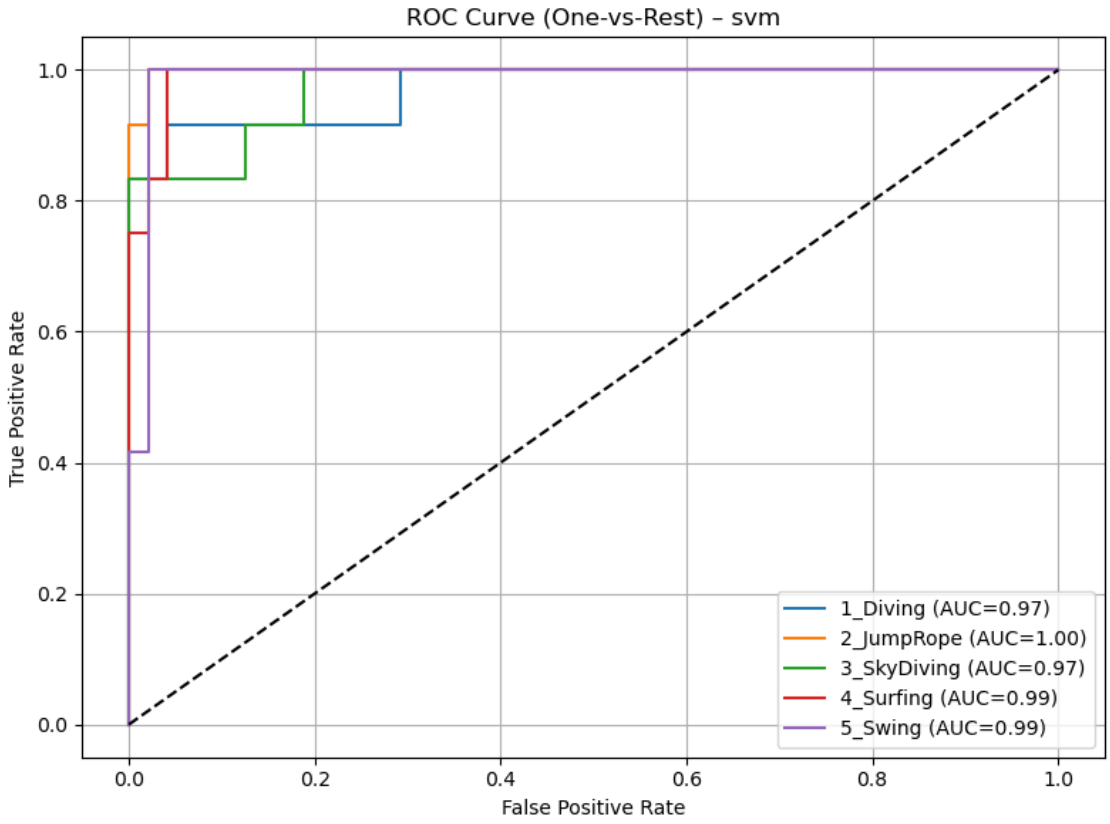
	Metric	Value
0	Accuracy	0.9167
1	Macro Precision	0.9203
2	Macro Recall	0.9167
3	Macro F1-Score	0.9157

Per-Class Classification Report

	precision	recall	f1-score	support
1_Diving	1.000000	0.833333	0.909091	12.000000
2_JumpRope	0.923077	1.000000	0.960000	12.000000
3_SkyDiving	0.909091	0.833333	0.869565	12.000000
4_Surfing	0.846154	0.916667	0.880000	12.000000

5_Swing	0.923077	1.000000	0.960000	12.000000
accuracy	0.916667	0.916667	0.916667	0.916667
macro avg	0.920280	0.916667	0.915731	60.000000
weighted avg	0.920280	0.916667	0.915731	60.000000






```
Evaluation completed successfully


SVM Evaluation Completed
```

[5]:
```python
# ==================================================
# 2  RANDOM FOREST CLASSIFIER
# ==================================================
# This block trains a Random Forest classifier and performs
# post-training feature analysis including:
# - Feature importance (tree-based)
# - t-SNE visualization
# - UMAP visualization
# ==================================================

print("\n------------------------------")
print("  Training Random Forest...")
print("------------------------------\n")

from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GridSearchCV
```

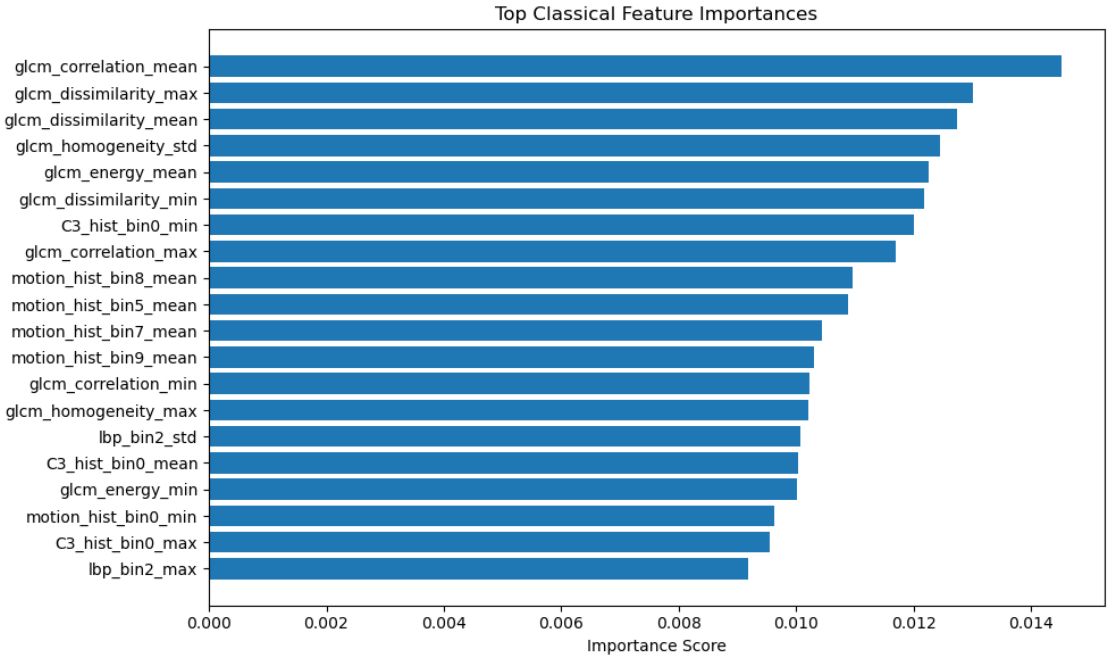
```
import time

from feature_extraction import (
    visualize_classical_feature_importance,
    visualize_classical_feature_space
)

# ------------------------------------------------
#   Step 1: Initialize Random Forest
# ------------------------------------------------
# Random Forest:
# - Does NOT require feature scaling
# - Learns feature importance via split criteria
# - Robust to noise and overfitting
# ------------------------------------------------

rf = RandomForestClassifier(
    random_state=RANDOM_STATE,
    n_jobs=-1,
)

# ------------------------------------------------
#   Step 2: Hyperparameter grid
# ------------------------------------------------
# Controls:
# - number of trees
# - tree depth
# - split granularity
# ------------------------------------------------

rf_param_grid = {
    "n_estimators": [100, 200],
    "max_depth": [10, 20, None],
    "min_samples_split": [2, 5],
}

# ------------------------------------------------
#   Step 3: GridSearchCV
# ------------------------------------------------
# Cross-validation ensures stable hyperparameter selection
# ------------------------------------------------

rf_grid = GridSearchCV(
    estimator=rf,
    param_grid=rf_param_grid,
    cv=CV_FOLDS,
    scoring="accuracy",
```

```
    n_jobs=-1,
    verbose=1,
)

# ------------------------------------------------
#   Step 4: Measure training time
# ------------------------------------------------

train_start_time = time.perf_counter()
rf_grid.fit(X_train, y_train)
train_end_time = time.perf_counter()

rf_training_time = train_end_time - train_start_time

print(f"\n  RF Training Time: {rf_training_time:.4f} seconds")
print(f"\n  Best RF Parameters: {rf_grid.best_params_}\n")

# ------------------------------------------------
#   Step 5: Extract best trained model
# ------------------------------------------------

best_rf = rf_grid.best_estimator_

# ================================================
#   CLASSICAL FEATURE ANALYSIS (POST-TRAINING)
# ================================================
# IMPORTANT:
# - Random Forest ALWAYS supports feature_importances_
# - Feature space visualization uses RAW features
#   (no scaling required for tree models)
# ================================================

# ------------------------------------------------
#   Step 6: Feature importance (Tree-based)
# ------------------------------------------------

feature_names = get_classical_feature_names(
    hist_bins=16,
    lbp_points=8,
    include_motion=True
)

visualize_classical_feature_importance(
    model=best_rf,
    feature_names=feature_names,
    top_k=20,
    save_name="rf_feature_importance.png"
```

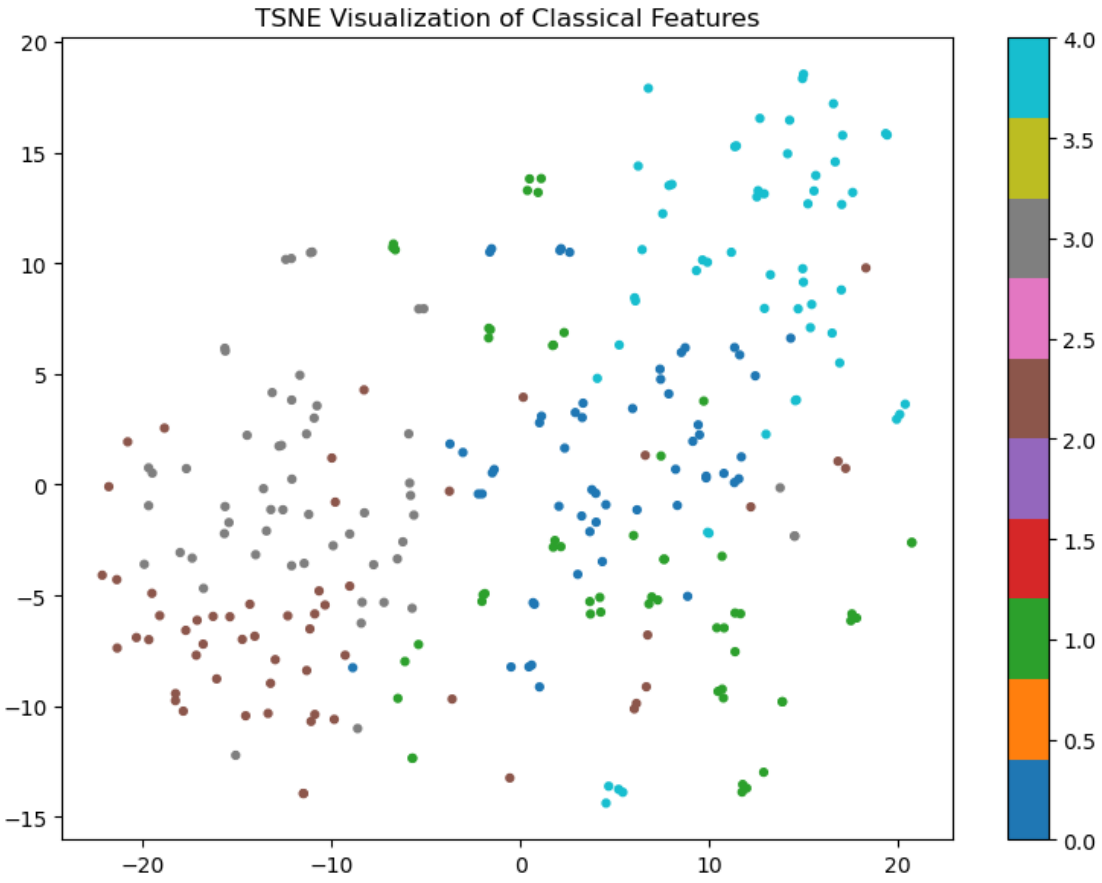
```
)

# ------------------------------------------------
#   Step 7: Feature space visualization (t-SNE)
# ------------------------------------------------
# Raw handcrafted features are sufficient for RF
# ------------------------------------------------

visualize_classical_feature_space(
    features=X_train,
    labels=y_train,
    method="tsne",
    save_name="rf_tsne.png"
)

# ------------------------------------------------
#   Step 8: Feature space visualization (UMAP)
# ------------------------------------------------

visualize_classical_feature_space(
    features=X_train,
    labels=y_train,
    method="umap",
    save_name="rf_umap.png"
)

# ================================================
#   INFERENCE TIME PER VIDEO
# ================================================
# Measures average prediction latency
# ================================================

inference_start_time = time.perf_counter()
y_pred = best_rf.predict(X_test)
inference_end_time = time.perf_counter()

rf_inference_time_per_video = (
    (inference_end_time - inference_start_time) / len(X_test)
)

print(
    f"\n  Avg Inference Time per Video: "
    f"{rf_inference_time_per_video:.6f} seconds\n"
)

# ================================================
# ROC / AUC PROBABILITIES
```

```python
# ================================================
# Random Forest natively supports predict_proba()
# ================================================

y_scores = best_rf.predict_proba(X_test)

# ================================================
# UNIFIED EVALUATION
# ================================================

class_names = [
    LABEL_TO_CLASS[i].replace("class_", "")
    for i in range(len(LABEL_TO_CLASS))
]

rf_metrics = evaluate_classification(
    y_true=y_test,
    y_pred=y_pred,
    y_scores=y_scores,
    class_names=class_names,
    model_name="random_forest"
)

# Store computational efficiency metrics
rf_metrics["training_time"] = rf_training_time
rf_metrics["inference_time_per_video"] = rf_inference_time_per_video

print("\n Random Forest Evaluation Completed")
```

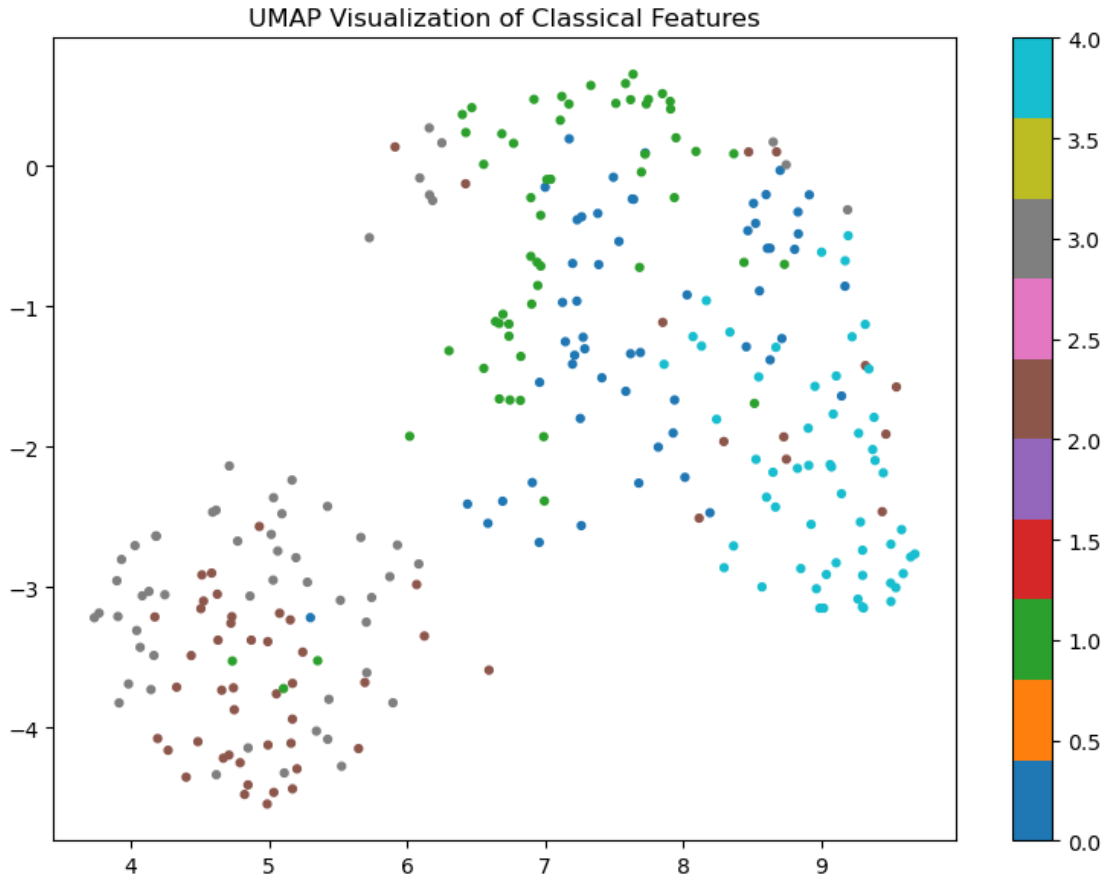
```
------------------------------
 Training Random Forest…
------------------------------

Fitting 5 folds for each of 12 candidates, totalling 60 fits

 RF Training Time: 6.6875 seconds

 Best RF Parameters: {'max_depth': 10, 'min_samples_split': 2, 'n_estimators':
200}

[Saved] Feature importance → /home/jovyan/va-assignment-
1/Student_2024ab05275_Video_Classification/results/feature_visualizations/classi
cal/rf_feature_importance.png
```



[Saved] TSNE plot → /home/jovyan/va-assignment-1/Student_2024ab05275_Video_Classification/results/feature_visualizations/classical/rf_tsne.png



[Saved] UMAP plot → /home/jovyan/va-assignment-1/Student_2024ab05275_Video_Classification/results/feature_visualizations/classical/rf_umap.png



Avg Inference Time per Video: 0.000797 seconds

Starting evaluation for: random_forest

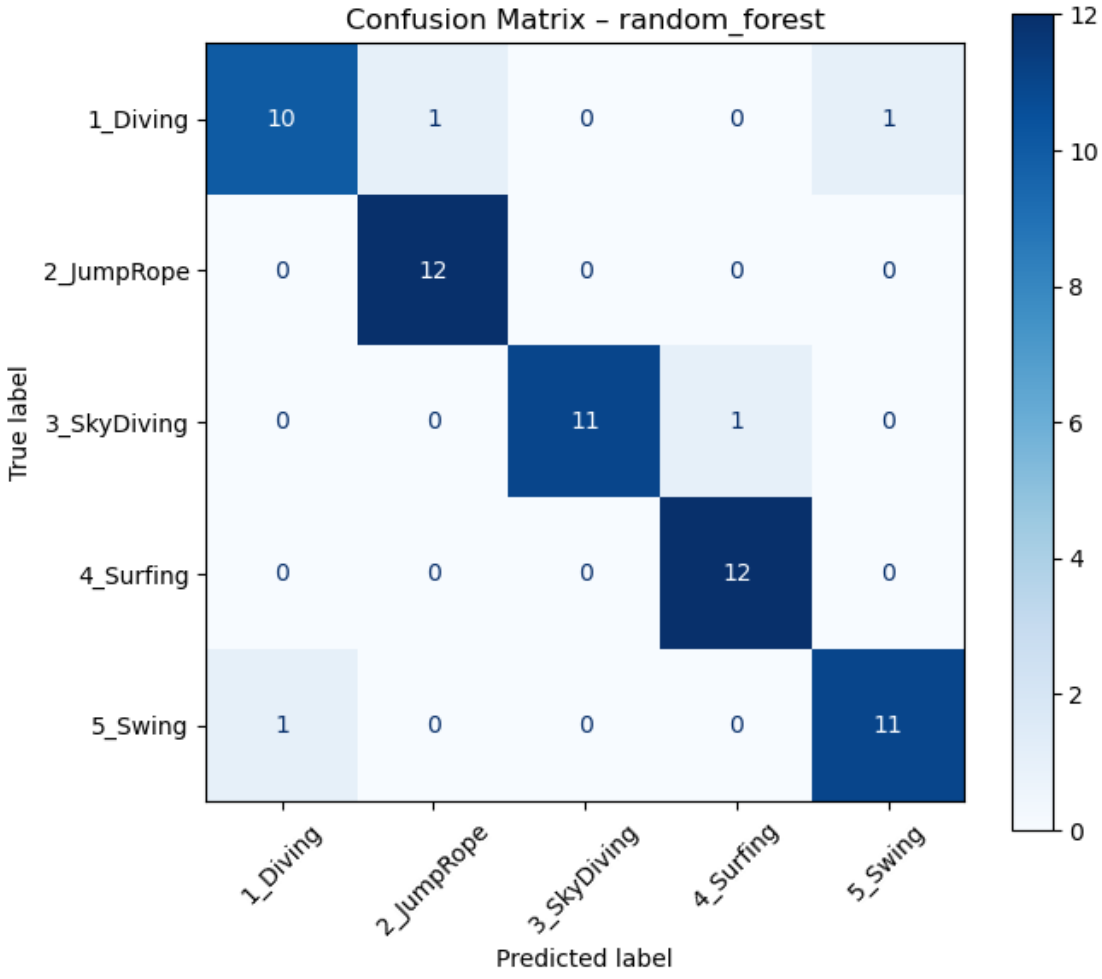
Overall Performance Metrics

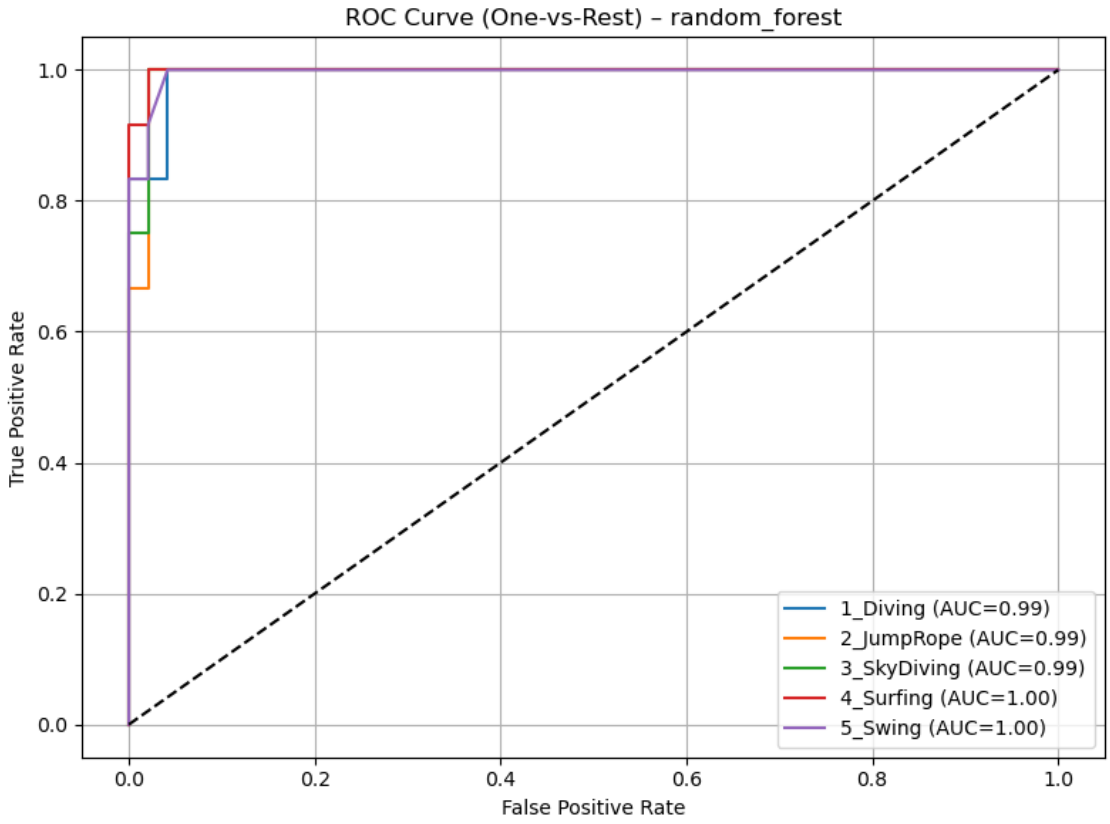
	Metric	Value
0	Accuracy	0.9333
1	Macro Precision	0.9344
2	Macro Recall	0.9333
3	Macro F1-Score	0.9326

Per-Class Classification Report

	precision	recall	f1-score	support
1_Diving	0.909091	0.833333	0.869565	12.000000
2_JumpRope	0.923077	1.000000	0.960000	12.000000
3_SkyDiving	1.000000	0.916667	0.956522	12.000000

4_Surfing	0.923077	1.000000	0.960000	12.000000
5_Swing	0.916667	0.916667	0.916667	12.000000
accuracy	0.933333	0.933333	0.933333	0.933333
macro avg	0.934382	0.933333	0.932551	60.000000
weighted avg	0.934382	0.933333	0.932551	60.000000






```
Evaluation completed successfully
```

```
Random Forest Evaluation Completed
```

[6]:
```
# ================================================
# 3  k-NEAREST NEIGHBORS (k-NN)
# ================================================
# This block trains a k-NN classifier and performs
# post-training feature analysis including:
# - t-SNE visualization
# - UMAP visualization
#
# NOTE:
# k-NN is a non-parametric, instance-based learner.
# It does NOT learn explicit feature weights.
# Therefore, feature importance is NOT applicable.
# ================================================

print("\n-------------------------------------")
print("  Training k-Nearest Neighbors...")
```

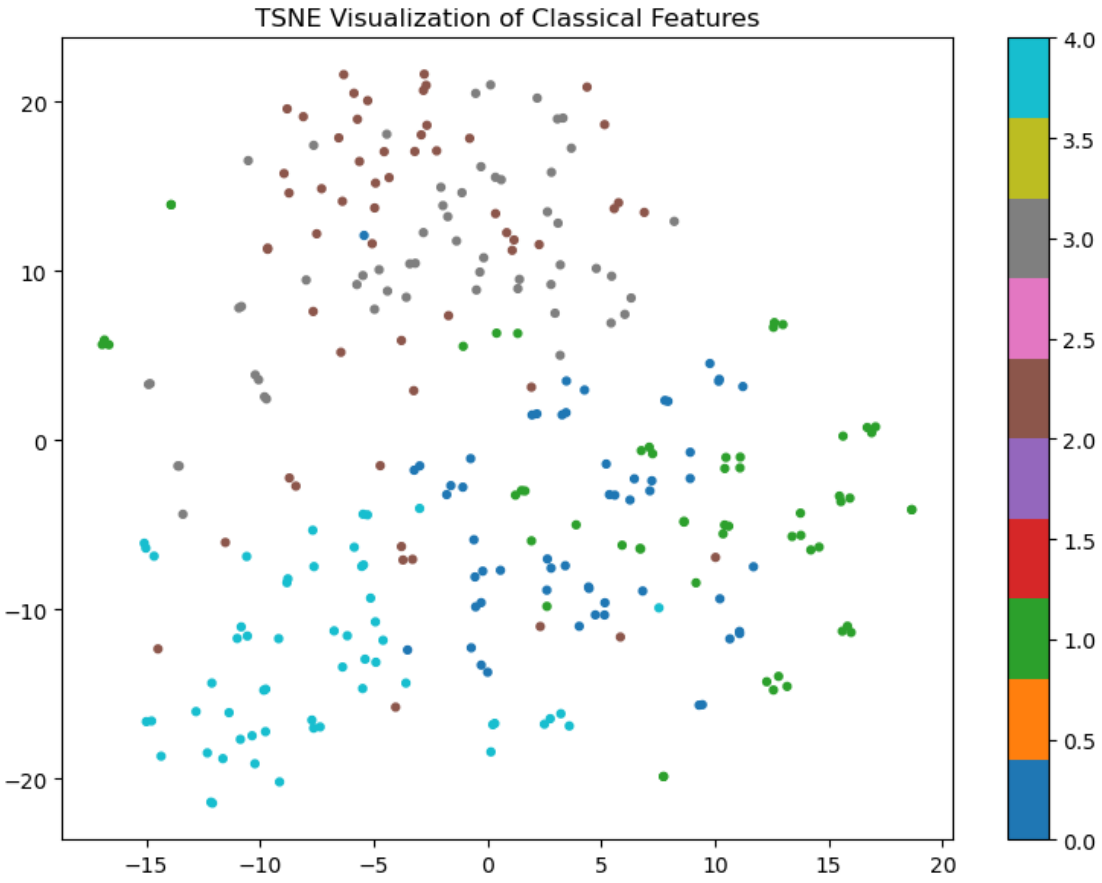
```
print("-----------------------------------\n")

# ------------------------------------------------
#   Step 1: Define k-NN pipeline
# ------------------------------------------------
# Feature scaling is CRITICAL for k-NN because:
# - Distance metrics are scale-sensitive
# - Unscaled features bias nearest neighbors
# ------------------------------------------------

knn_pipeline = Pipeline(
    steps=[
        ("scaler", StandardScaler()),
        (
            "knn",
            KNeighborsClassifier(
                weights="distance"   # improves probability estimates
            ),
        ),
    ]
)

# ------------------------------------------------
#   Step 2: Hyperparameter grid
# ------------------------------------------------
# - n_neighbors controls local vs global behavior
# - metric defines distance computation
# ------------------------------------------------

knn_param_grid = {
    "knn__n_neighbors": [3, 5, 7, 9],
    "knn__metric": ["euclidean", "manhattan"],
}

# ------------------------------------------------
#   Step 3: GridSearchCV
# ------------------------------------------------
# Cross-validation ensures robust selection of k
# ------------------------------------------------

knn_grid = GridSearchCV(
    estimator=knn_pipeline,
    param_grid=knn_param_grid,
    cv=CV_FOLDS,
    scoring="accuracy",
    n_jobs=-1,
    verbose=1,
```

```
)

# ------------------------------------------------
#   Step 4: Measure training time
# ------------------------------------------------
# k-NN has minimal "training" time (lazy learner)
# ------------------------------------------------

train_start_time = time.perf_counter()
knn_grid.fit(X_train, y_train)
train_end_time = time.perf_counter()

knn_training_time = train_end_time - train_start_time

print(f"\n  k-NN Training Time: {knn_training_time:.4f} seconds")
print(f"  Best k-NN Parameters: {knn_grid.best_params_} \n")

# ------------------------------------------------
#   Step 5: Extract best trained model
# ------------------------------------------------
# best_knn is a full pipeline:
#   scaler → k-NN
# ------------------------------------------------

best_knn = knn_grid.best_estimator_

# ================================================
#   FEATURE SPACE VISUALIZATION (POST-TRAINING)
# ================================================
# IMPORTANT:
# Feature space MUST match the scaled representation
# used internally by k-NN
# ================================================

# ------------------------------------------------
#   Step 6: Extract scaled training features
# ------------------------------------------------

X_train_scaled = best_knn.named_steps["scaler"].transform(X_train)

# ------------------------------------------------
#   Step 7: t-SNE visualization
# ------------------------------------------------

visualize_classical_feature_space(
    features=X_train_scaled,
    labels=y_train,
```

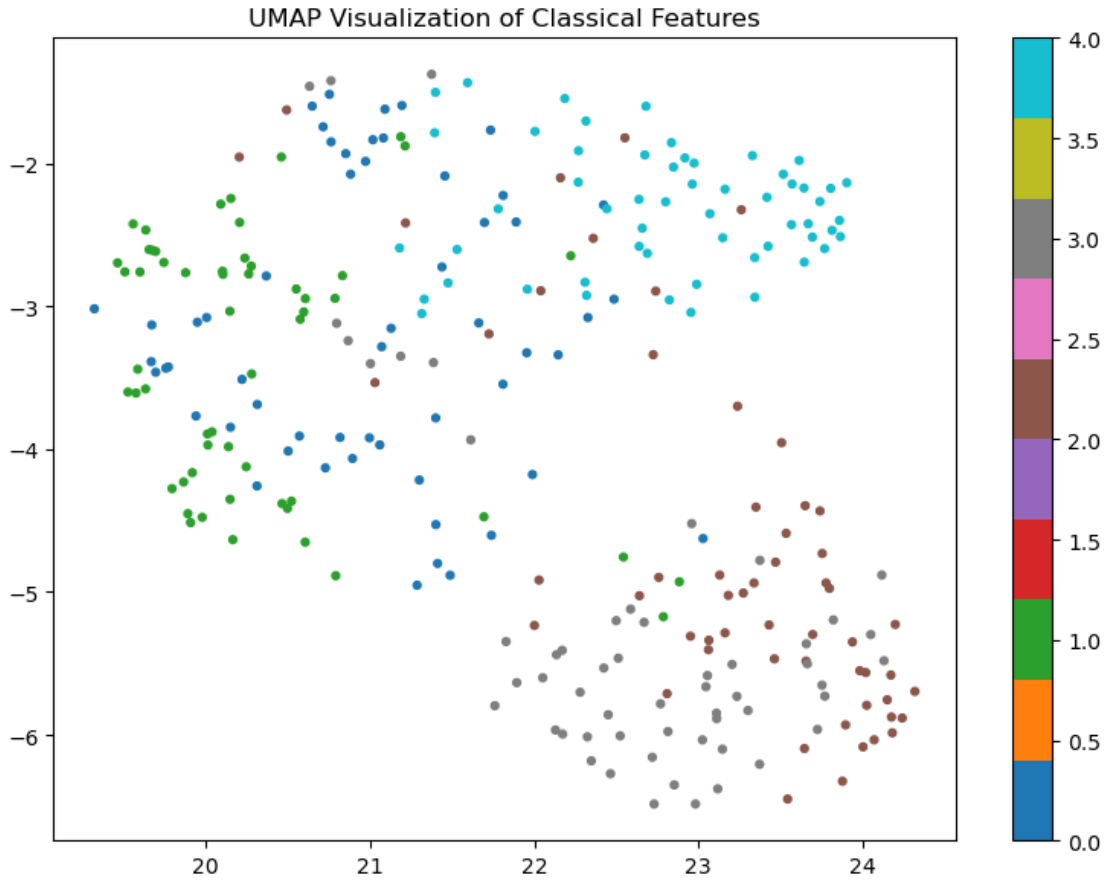
```
    method="tsne",
    save_name="knn_tsne.png"
)

# ------------------------------------------------
#   Step 8: UMAP visualization
# ------------------------------------------------

visualize_classical_feature_space(
    features=X_train_scaled,
    labels=y_train,
    method="umap",
    save_name="knn_umap.png"
)

# ================================================
#   INFERENCE TIME PER VIDEO
# ================================================
# k-NN inference is slower due to distance computations
# ================================================

inference_start_time = time.perf_counter()
y_pred = best_knn.predict(X_test)
inference_end_time = time.perf_counter()

knn_inference_time_per_video = (
    (inference_end_time - inference_start_time) / len(X_test)
)

print(
    f"  Avg Inference Time per Video: "
    f"{knn_inference_time_per_video:.6f} seconds\n"
)

# ================================================
# ROC / AUC PROBABILITIES
# ================================================
# weights="distance" enables predict_proba()
# ================================================

y_scores = best_knn.predict_proba(X_test)

# ================================================
# UNIFIED EVALUATION
# ================================================

class_names = [
```

```
    LABEL_TO_CLASS[i].replace("class_", "")
    for i in range(len(LABEL_TO_CLASS))
]

knn_metrics = evaluate_classification(
    y_true=y_test,
    y_pred=y_pred,
    y_scores=y_scores,
    class_names=class_names,
    model_name="knn",
)

# Store computational efficiency metrics
knn_metrics["training_time"] = knn_training_time
knn_metrics["inference_time_per_video"] = knn_inference_time_per_video

print("\n  k-NN Evaluation Completed")
```

```
------------------------------------
  Training k-Nearest Neighbors…
------------------------------------

Fitting 5 folds for each of 8 candidates, totalling 40 fits

  k-NN Training Time: 0.1003 seconds
  Best k-NN Parameters: {'knn__metric': 'manhattan', 'knn__n_neighbors': 7}

[Saved] TSNE plot → /home/jovyan/va-assignment-
1/Student_2024ab05275_Video_Classification/results/feature_visualizations/classi
cal/knn_tsne.png
```



[Saved] UMAP plot → /home/jovyan/va-assignment-1/Student_2024ab05275_Video_Classification/results/feature_visualizations/classical/knn_umap.png



Avg Inference Time per Video: 0.000079 seconds

Starting evaluation for: knn

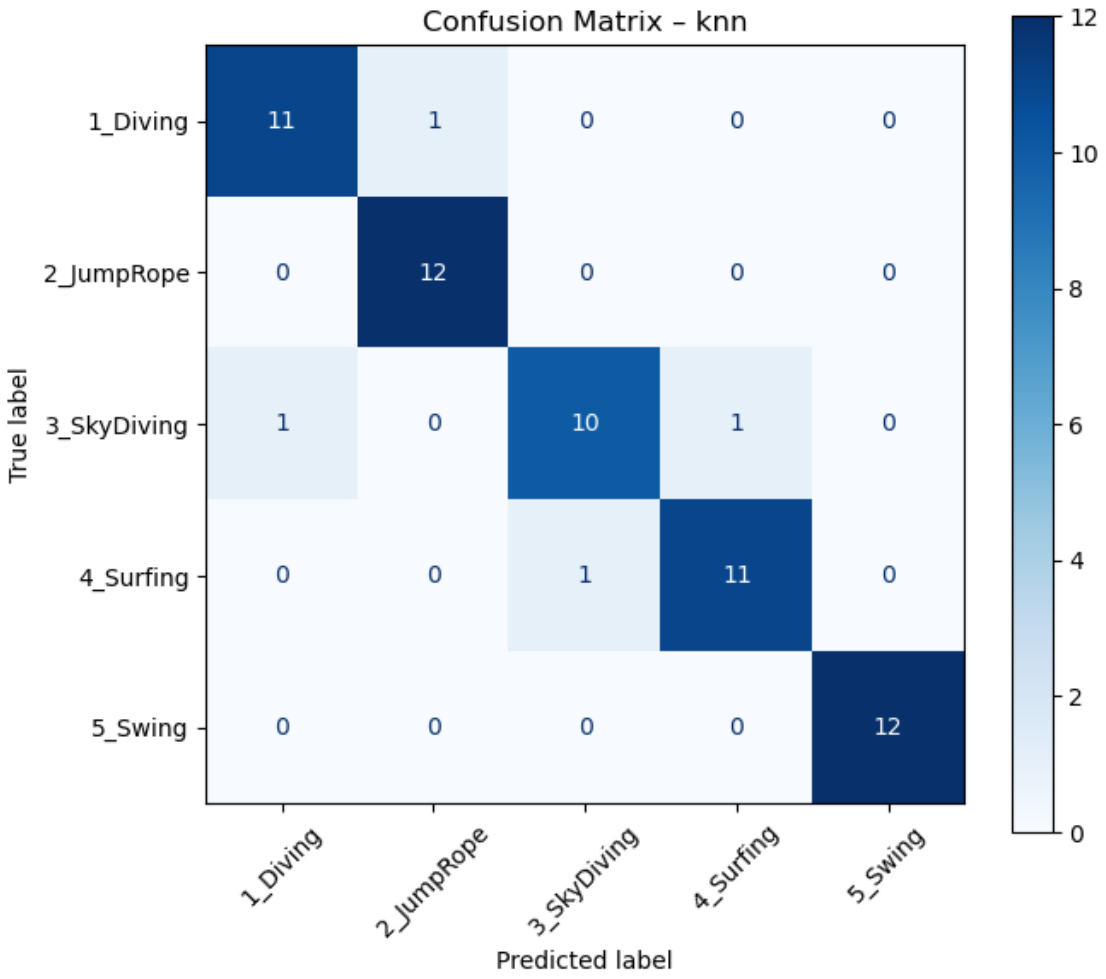
Overall Performance Metrics

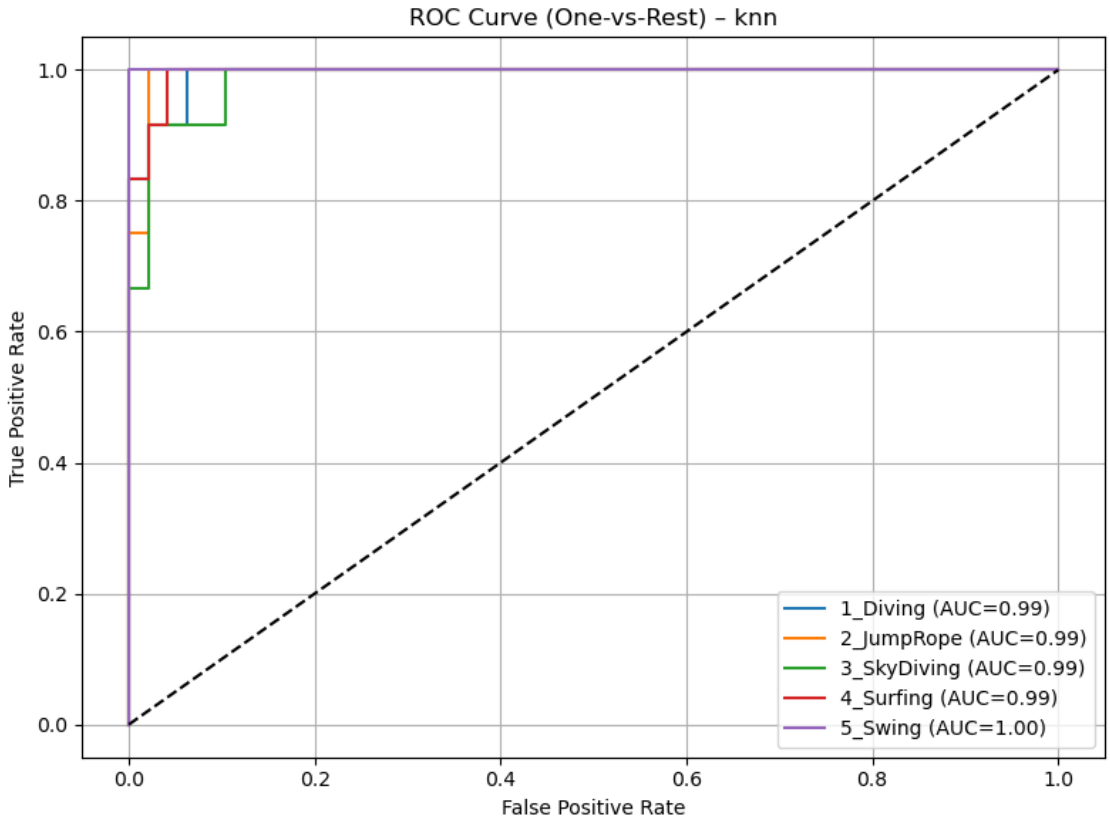
	Metric	Value
0	Accuracy	0.9333
1	Macro Precision	0.9331
2	Macro Recall	0.9333
3	Macro F1-Score	0.9326

Per-Class Classification Report

	precision	recall	f1-score	support
1_Diving	0.916667	0.916667	0.916667	12.000000
2_JumpRope	0.923077	1.000000	0.960000	12.000000
3_SkyDiving	0.909091	0.833333	0.869565	12.000000
4_Surfing	0.916667	0.916667	0.916667	12.000000

5_Swing	1.000000	1.000000	1.000000	12.000000
accuracy	0.933333	0.933333	0.933333	0.933333
macro avg	0.933100	0.933333	0.932580	60.000000
weighted avg	0.933100	0.933333	0.932580	60.000000






```
Evaluation completed successfully


k-NN Evaluation Completed
```

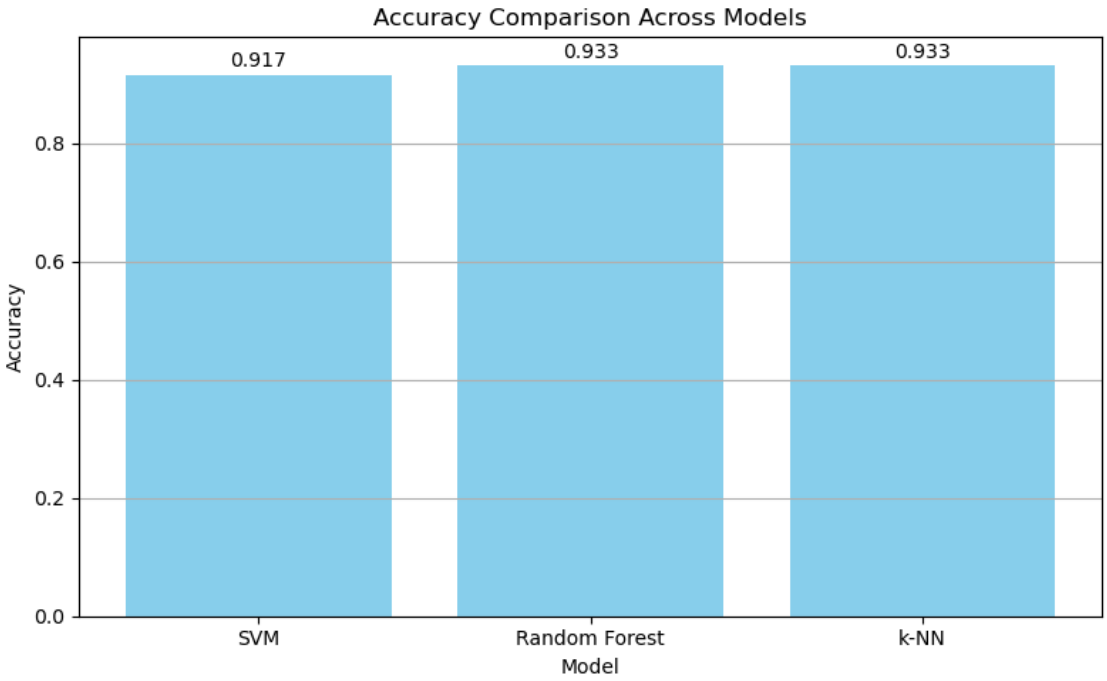
[7]:
```python
# ================================================
#   Comparative Analysis of All Three ML Models
# ================================================

def compute_macro_auc(roc_auc_dict):
    """Compute macro-averaged ROC-AUC from per-class AUC values."""
    return sum(roc_auc_dict.values()) / len(roc_auc_dict)

# ----------------------------
# 1  Build formatted comparison table
# ----------------------------
comparison_df = pd.DataFrame([
    {
        "Model": "SVM",
        "Accuracy": round(svm_metrics["accuracy"], 4),
        "Macro Precision": round(svm_metrics["precision_macro"], 4),
```

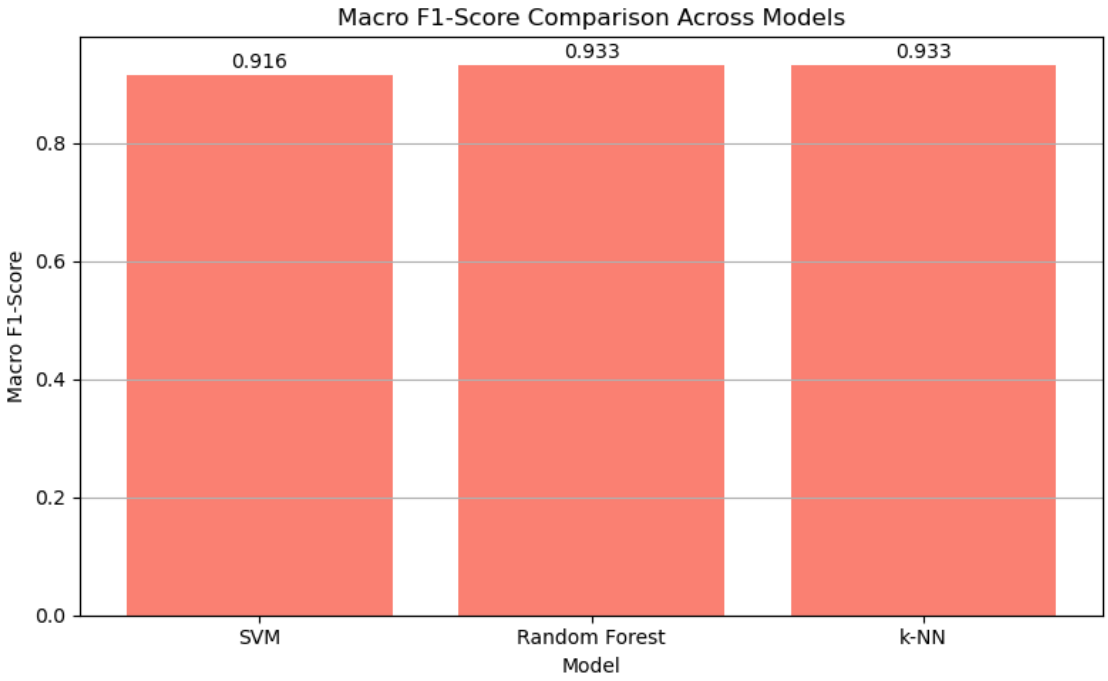
```
        "Macro Recall": round(svm_metrics["recall_macro"], 4),
        "Macro F1-Score": round(svm_metrics["f1_macro"], 4),
        "Macro ROC-AUC": round(compute_macro_auc(svm_metrics["roc_auc"]), 4),
    },
    {
        "Model": "Random Forest",
        "Accuracy": round(rf_metrics["accuracy"], 4),
        "Macro Precision": round(rf_metrics["precision_macro"], 4),
        "Macro Recall": round(rf_metrics["recall_macro"], 4),
        "Macro F1-Score": round(rf_metrics["f1_macro"], 4),
        "Macro ROC-AUC": round(compute_macro_auc(rf_metrics["roc_auc"]), 4),
    },
    {
        "Model": "k-NN",
        "Accuracy": round(knn_metrics["accuracy"], 4),
        "Macro Precision": round(knn_metrics["precision_macro"], 4),
        "Macro Recall": round(knn_metrics["recall_macro"], 4),
        "Macro F1-Score": round(knn_metrics["f1_macro"], 4),
        "Macro ROC-AUC": round(compute_macro_auc(knn_metrics["roc_auc"]), 4),
    },
])

# Display table
print("\n Comparative Performance Metrics")
display(comparison_df)

# ----------------------------
# Helper function for annotated bar plots
# ----------------------------
def plot_metric_bar(df, metric, title, color="skyblue"):
    plt.figure(figsize=(8, 5))
    bars = plt.bar(df["Model"], df[metric], color=color)
    plt.xlabel("Model")
    plt.ylabel(metric)
    plt.title(title)
    plt.grid(axis="y")

    # Annotate bars
    for bar in bars:
        yval = bar.get_height()
        plt.text(bar.get_x() + bar.get_width()/2, yval + 0.005, f"{yval:.3f}",␣
 ↪ha="center", va="bottom")

    plt.tight_layout()
    plt.show()

# ----------------------------
```

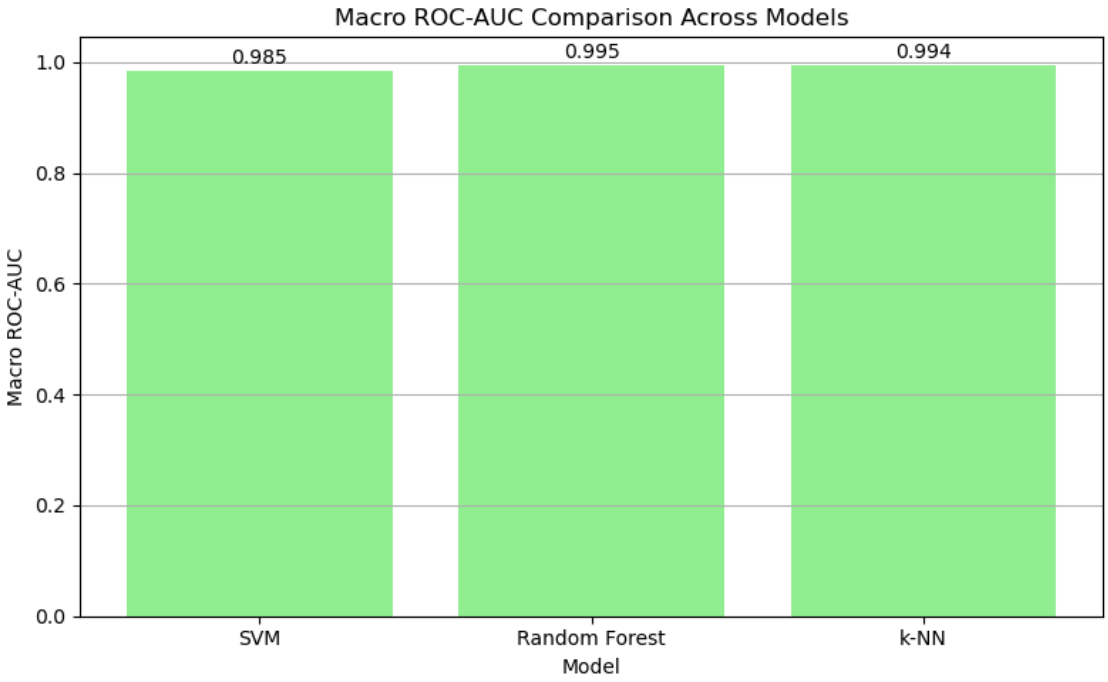
```
# 2 Individual Metric Charts (keep old charts)
# ----------------------------
plot_metric_bar(comparison_df, "Accuracy", "Accuracy Comparison Across Models",␣
 ↪color="skyblue")
plot_metric_bar(comparison_df, "Macro F1-Score", "Macro F1-Score Comparison␣
 ↪Across Models", color="salmon")
plot_metric_bar(comparison_df, "Macro ROC-AUC", "Macro ROC-AUC Comparison␣
 ↪Across Models", color="lightgreen")

# ----------------------------
# 3 Single Grouped Bar Chart
# ----------------------------
metrics = ["Accuracy", "Macro Precision", "Macro Recall", "Macro F1-Score",␣
 ↪"Macro ROC-AUC"]

# Extract values for each model
svm_values = [comparison_df.loc[comparison_df["Model"]=="SVM", m].values[0] for␣
 ↪m in metrics]
rf_values = [comparison_df.loc[comparison_df["Model"]=="Random Forest", m].
 ↪values[0] for m in metrics]
knn_values = [comparison_df.loc[comparison_df["Model"]=="k-NN", m].values[0]␣
 ↪for m in metrics]

x = np.arange(len(metrics))  # metric positions
width = 0.25

plt.figure(figsize=(12, 6))
plt.bar(x - width, svm_values, width, label="SVM", color="skyblue")
plt.bar(x, rf_values, width, label="Random Forest", color="salmon")
plt.bar(x + width, knn_values, width, label="k-NN", color="lightgreen")

# Annotate bars
for i, vals in enumerate([svm_values, rf_values, knn_values]):
    for j, val in enumerate(vals):
        xpos = x[j] - width if i==0 else x[j] if i==1 else x[j] + width
        plt.text(xpos, val + 0.005, f"{val:.3f}", ha="center", va="bottom",␣
 ↪fontsize=9)

plt.xticks(x, metrics, rotation=25)
plt.ylabel("Score")
plt.title("Comparative Performance of ML Models (Grouped)")
plt.ylim(0, 1.05)
plt.legend(loc='upper left', bbox_to_anchor=(1, 1))  # x=1 (right), y=0 (bottom)
plt.grid(axis="y", linestyle="--", alpha=0.7)
plt.tight_layout()
plt.show()
```

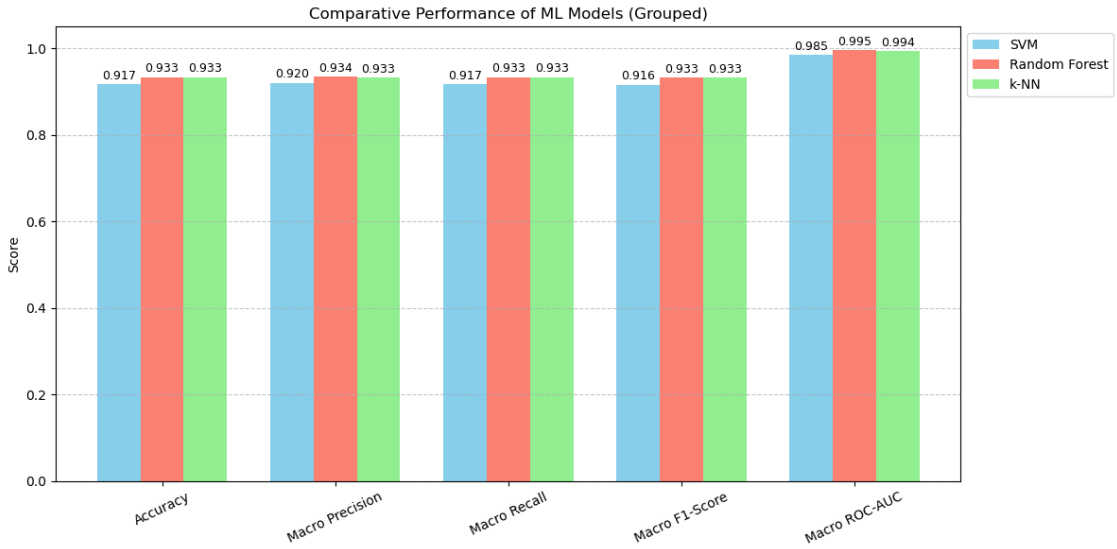
```
print("\n  Comparative analysis completed successfully.")
```

```
  Comparative Performance Metrics

           Model  Accuracy  Macro Precision  Macro Recall  Macro F1-Score  \
0            SVM    0.9167           0.9203        0.9167          0.9157
1  Random Forest    0.9333           0.9344        0.9333          0.9326
2           k-NN    0.9333           0.9331        0.9333          0.9326

   Macro ROC-AUC
0         0.9847
1         0.9950
2         0.9938
```




Macro F1-Score Comparison Across Models
0.916
0.933
0.933
Macro F1-Score
0.0
0.2
0.4
0.6
0.8
SVM
Random Forest
k-NN
Model


Macro ROC-AUC Comparison Across Models
0.985
0.995
0.994
Macro ROC-AUC
0.0
0.2
0.4
0.6
0.8
1.0
SVM
Random Forest
k-NN
Model



Comparative analysis completed successfully.

[8]:
```
# ============================================================
# CLASSICAL ML - FULL COMPARATIVE REPORT (DISPLAY + SAVE)
# ============================================================
# ============================================================
# 1  GENERATE FULL REPORT (NOTEBOOK DISPLAY)
# ============================================================

def generate_full_report(svm_metrics, rf_metrics, knn_metrics, class_names):
    """
    Generates a fully PDF-safe comparative report
    for classical ML models (SVM, Random Forest, k-NN).

      Renders correctly in:
       - Jupyter Notebook
       - Docker
       - nbconvert PDF
    """

    models = {
        "SVM": svm_metrics,
        "Random Forest": rf_metrics,
        "k-NN": knn_metrics
    }

    # ------------------------------------------------------
    # Helper: Compute Macro ROC-AUC
```

```
    # ----------------------------------------------------
    def compute_macro_auc(roc_auc_dict):
        return sum(roc_auc_dict.values()) / len(roc_auc_dict)

    # ----------------------------------------------------
    # 1  Overall Performance Metrics
    # ----------------------------------------------------
    overall_df = pd.DataFrame([
        {
            "Model": m,
            "Accuracy": round(v["accuracy"], 4),
            "Macro Precision": round(v["precision_macro"], 4),
            "Macro Recall": round(v["recall_macro"], 4),
            "Macro F1-Score": round(v["f1_macro"], 4),
            "Macro ROC-AUC": round(compute_macro_auc(v["roc_auc"]), 4),
        }
        for m, v in models.items()
    ])

    display(Markdown("##  Overall Performance Metrics"))
    display(overall_df)

    # ----------------------------------------------------
    # 2  Per-Class Metrics
    # ----------------------------------------------------
    for model_name, metrics in models.items():
        display(Markdown(f"###  Per-Class Metrics - {model_name}"))

        class_df = pd.DataFrame(metrics["classification_report"]).T

        rows_order = (
            [c for c in class_names if c in class_df.index] +
            [r for r in ["accuracy", "macro avg", "weighted avg"] if r in␣
↪class_df.index]
        )

        class_df = class_df.loc[rows_order]

        display(class_df.round(4))

    # ----------------------------------------------------
    # 3  Key Observations
    # ----------------------------------------------------
    accuracy = {m: v["accuracy"] for m, v in models.items()}
    macro_f1 = {m: v["f1_macro"] for m, v in models.items()}
    macro_precision = {m: v["precision_macro"] for m, v in models.items()}
    macro_recall = {m: v["recall_macro"] for m, v in models.items()}
```

```
    best_acc_model = max(accuracy, key=accuracy.get)
    best_f1_model = max(macro_f1, key=macro_f1.get)

    observations_df = pd.DataFrame([
        ["Highest Accuracy", f"{best_acc_model} ({accuracy[best_acc_model]:.
↪3f})"],
        ["Best Overall Performance (Macro F1-Score)", f"{best_f1_model}␣
↪({macro_f1[best_f1_model]:.3f})"],
        ["Precision-Recall Balance",
         f"Random Forest: Precision={macro_precision['Random Forest']:.3f}, "
         f"Recall={macro_recall['Random Forest']:.3f}"],
        ["Model Behavior",
         "SVM: strong margin-based classifier (probability calibration␣
↪needed)\n"
         "k-NN: sensitive to feature scaling and local neighborhoods"]
    ], columns=["Observation", "Details"])

    display(Markdown("##  Key Observations"))
    display(observations_df)

    # ------------------------------------------------------
    # 4 Model Ranking
    # ------------------------------------------------------
    ranking_df = overall_df.copy()

    metric_cols = [
        "Accuracy",
        "Macro Precision",
        "Macro Recall",
        "Macro F1-Score",
        "Macro ROC-AUC"
    ]

    for col in metric_cols:
        ranking_df[f"{col} Rank"] = ranking_df[col].rank(
            ascending=False, method="min"
        ).astype(int)

    display(Markdown("##  Model Ranking by Metric (1 = Best)"))
    display(ranking_df)

    # ------------------------------------------------------
    # 5 Conclusion
    # ------------------------------------------------------
    display(Markdown(
        """
```

```
###  Final Conclusion

Random Forest demonstrates the most consistent and reliable performance
across accuracy, precision, recall, and F1-score, making it the most suitable
classical machine learning model for multi-class video action recognition
in this study.
"""
    ))



# ============================================================
# 2  SAVE FULL REPORT (SINGLE CSV EXPORT)
# ============================================================

def save_comparative_report(
    svm_metrics,
    rf_metrics,
    knn_metrics,
    save_path=None
):
    """
    Saves classical ML metrics in EXACT SAME FORMAT
    as Deep Learning comparison CSV.

    This guarantees:
    - Direct CSV comparison
    - Unified plots
    - Automated report generation
    """

    models = {
        "SVM": svm_metrics,
        "Random Forest": rf_metrics,
        "k-NN": knn_metrics
    }

    # ------------------------------------------------
    # 1  Build metrics table (STRICT COLUMN ORDER)
    # ------------------------------------------------
    df = pd.DataFrame([
        {
            "Model": model_name,
            "Accuracy": round(m["accuracy"], 6),
            "Precision (Macro)": round(m["precision_macro"], 6),
            "Recall (Macro)": round(m["recall_macro"], 6),
            "F1-score (Macro)": round(m["f1_macro"], 6),
```

```
            "Training Time (s)": round(m["training_time"], 6),
            "Inference Time / Sample (s)": round(
                m["inference_time_per_video"], 6
            )
        }
        for model_name, m in models.items()
    ])

    # ------------------------------------------------
    # 2 Ranking logic (MATCH DL BEHAVIOR)
    # ------------------------------------------------
    higher_better = [
        "Accuracy",
        "Precision (Macro)",
        "Recall (Macro)",
        "F1-score (Macro)"
    ]

    lower_better = [
        "Training Time (s)",
        "Inference Time / Sample (s)"
    ]

    for col in higher_better:
        df[f"{col} Rank"] = df[col].rank(
            ascending=False,
            method="min"
        ).astype(int)

    for col in lower_better:
        df[f"{col} Rank"] = df[col].rank(
            ascending=True,
            method="min"
        ).astype(int)

    # ------------------------------------------------
    # 3 Enforce FINAL COLUMN ORDER (CRITICAL)
    # ------------------------------------------------
    final_columns = [
        "Model",
        "Accuracy",
        "Precision (Macro)",
        "Recall (Macro)",
        "F1-score (Macro)",
        "Training Time (s)",
        "Inference Time / Sample (s)",
        "Accuracy Rank",
```

```
        "Precision (Macro) Rank",
        "Recall (Macro) Rank",
        "F1-score (Macro) Rank",
        "Training Time (s) Rank",
        "Inference Time / Sample (s) Rank"
    ]

    df = df[final_columns]

    # ------------------------------------------------
    # 4  Save CSV
    # ------------------------------------------------
    os.makedirs(os.path.dirname(save_path), exist_ok=True)
    df.to_csv(save_path, index=False)

    print(f"\n  Classical ML comparison saved → {save_path}")



# ============================================================
# 3  USAGE
# ============================================================

# Display report
generate_full_report(
    svm_metrics,
    rf_metrics,
    knn_metrics,
    class_names
)

# Save report

# Current file is inside /code
CODE_DIR = Path.cwd()

# Project root is one level above
PROJECT_ROOT = CODE_DIR.parent

save_comparative_report(
    svm_metrics,
    rf_metrics,
    knn_metrics,
    save_path=f"{PROJECT_ROOT}/results/classical_comparison.csv"
)
```

1.1 Overall Performance Metrics

	Model	Accuracy	Macro Precision	Macro Recall	Macro F1-Score	Macro ROC-AUC
0	SVM	0.9167	0.9203	0.9167	0.9157	0.9847
1	Random Forest	0.9333	0.9344	0.9333	0.9326	0.9950
2	k-NN	0.9333	0.9331	0.9333	0.9326	0.9938

1.1.1 Per-Class Metrics — SVM

	precision	recall	f1-score	support
1_Diving	1.0000	0.8333	0.9091	12.0000
2_JumpRope	0.9231	1.0000	0.9600	12.0000
3_SkyDiving	0.9091	0.8333	0.8696	12.0000
4_Surfing	0.8462	0.9167	0.8800	12.0000
5_Swing	0.9231	1.0000	0.9600	12.0000
accuracy	0.9167	0.9167	0.9167	0.9167
macro avg	0.9203	0.9167	0.9157	60.0000
weighted avg	0.9203	0.9167	0.9157	60.0000

1.1.2 Per-Class Metrics — Random Forest

	precision	recall	f1-score	support
1_Diving	0.9091	0.8333	0.8696	12.0000
2_JumpRope	0.9231	1.0000	0.9600	12.0000
3_SkyDiving	1.0000	0.9167	0.9565	12.0000
4_Surfing	0.9231	1.0000	0.9600	12.0000
5_Swing	0.9167	0.9167	0.9167	12.0000
accuracy	0.9333	0.9333	0.9333	0.9333
macro avg	0.9344	0.9333	0.9326	60.0000
weighted avg	0.9344	0.9333	0.9326	60.0000

1.1.3 Per-Class Metrics — k-NN

	precision	recall	f1-score	support
1_Diving	0.9167	0.9167	0.9167	12.0000
2_JumpRope	0.9231	1.0000	0.9600	12.0000
3_SkyDiving	0.9091	0.8333	0.8696	12.0000
4_Surfing	0.9167	0.9167	0.9167	12.0000
5_Swing	1.0000	1.0000	1.0000	12.0000
accuracy	0.9333	0.9333	0.9333	0.9333
macro avg	0.9331	0.9333	0.9326	60.0000
weighted avg	0.9331	0.9333	0.9326	60.0000

1.2 Key Observations

```
                              Observation  \
0                          Highest Accuracy
1  Best Overall Performance (Macro F1-Score)
2                   Precision-Recall Balance
3                             Model Behavior

                                        Details
0                          Random Forest (0.933)
1                                   k-NN (0.933)
2      Random Forest: Precision=0.934, Recall=0.933
3  SVM: strong margin-based classifier (probabili…
```

1.3 Model Ranking by Metric (1 = Best)

```
           Model  Accuracy  Macro Precision  Macro Recall  Macro F1-Score  \
0            SVM    0.9167           0.9203        0.9167          0.9157
1  Random Forest    0.9333           0.9344        0.9333          0.9326
2           k-NN    0.9333           0.9331        0.9333          0.9326

   Macro ROC-AUC  Accuracy Rank  Macro Precision Rank  Macro Recall Rank  \
0         0.9847              3                     3                  3
1         0.9950              1                     1                  1
2         0.9938              1                     2                  1

   Macro F1-Score Rank  Macro ROC-AUC Rank
0                    3                   3
1                    1                   1
2                    1                   2
```

1.3.1 Final Conclusion

Random Forest demonstrates the most consistent and reliable performance across accuracy, precision, recall, and F1-score, making it the most suitable classical machine learning model for multi-class video action recognition in this study.

```
 Classical ML comparison saved → /home/jovyan/va-assignment-
1/Student_2024ab05275_Video_Classification/results/classical_comparison.csv
```

[ ]: